Phase Manifold Model:

First Principles and Mathematics

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Time is angular, space is emergent relational motion, depth is different, and c is stillness.

Abstract

The Phase Manifold Model (PMM) begins with the event itself. Possibility enters realized interaction, and completed interaction becomes the history from which later events develop. Before, During, and After describe this continuous conversion. One complete shared event is a Field Transition Unit:

$$\mathrm{F}_{tu} \;=\; B \circ D \circ A$$

Before carries possibility without realized space. Contact @ opens During, where the two internal vantages realize spatial separation and motion. The terminal anti-contact $-@$ closes that activity into After, retaining the event as physical history. Time is angular. Depth arises from the relation between D vantages. Space and motion are expressions of one realized relational process. The c-order ladder $c^0 \to c^1 \to c^2 \to c^3$ tracks that complete passage.

Each elementary pairwise interaction is one FTU. Its completed event produces two lineage-endpoint A records with one shared event identity and one event-level source energy. Active depth ends at transcription, while the completed x-y readout and phase history remain available as record. The three phase-time readouts retain their branch-dependent rank and conditional physical interpretations; their temporal metric remains open.

The single-FTU vantage follows one event from potential through realized geometry to completed record. The multi-FTU vantage follows the relations among events: completed records accumulate in Φ, influence subsequent contact, and condition relational clock response. The Angular Temporal Recursion Principle joins these vantages through $c^3 \to \Phi \to c^0$. Within coherent channels, the angular relations among contributions supply the interference construction. Action supplies a common measure across B, D, and A.

PMM IN ONE PAGE

Central proposition. Physical reality is an event process. Before carries possibility. During realizes spatial relationships. After writes what occurred into persistent history. That history participates in the possibility of what happens next.

PMM follows this process at two scales: inside one complete FTU and across the relationships of many FTUs. The Ladder of Light orders each event; angular progression governs its realized passage. The accumulated record Φ connects completed events to their successors.

$$B \;\to\; D \;\to\; A$$

Before (B) carries pre-spatial possibility. **During (D)** carries active realization and spatial geometry. **After (A)** carries the completed post-spatial record. PMM's human-facing correspondences are B $\to$ Wave, D $\to$ Particle, and A $\to$ Cube. One complete event is a Field Transition Unit:

$$\mathrm{F}_{tu} \;=\; B \circ D \circ A$$

The FTU physical c-order representation is:

$$B \;=\; (c^0)_{mp}$$

$$D \;=\; (c^1, c^2)c$$

$$A \;=\; (c^3)k_B$$

With its complete representation as:

$$\mathrm{F}^{c}_{tu} \;=\; ((c^0)_{mp}, (c^1, c^2)c, (c^3)k_B)$$

The event-grade progression is:

$$c^0 \ \rightarrow \ c^1 \ \rightarrow \ c^2 \ \rightarrow \ c^3$$

The constants act as phase creases: $m_{_p}$ anchors entry, c scales realization, and k_B marks transcription into information units. Their standard physical values are accepted inputs; PMM's claim is their assignment to the event architecture. The native ladder ends at c^3.

During D, common scale expands and vantage-scale imbalance contracts:

$$L \ = \ \ell_E \theta^D$$

$$r \ = \ r_0 \ \cos(\theta^D/2)$$

The mode b describes planar-shape disagreement between the two vantages. At terminal closure, r = b = Σ = 0; b's evolution remains candidate/open. Internal depth z^D is distinct from inter-participant distance δ and the traversal readout d.

M_{op} — **Motion Operations Principle:** Space is emergent relational motion. B carries potential relation and potential motion, D realizes relation as active space and relational motion, and A retains the completed relation as persistent record. The realization crease c is the invariant still reference of D.

The comparison-time kinematic readout is $M_{op}(\delta_{ij}) \equiv (\delta_{ij},\, d\delta_{ij}/dt,\, d^2\delta_{ij}/dt^2)$, with t the comparison time. It expresses relational separation, motion, and acceleration; the dynamics are developed separately in §20.2.

The time readout is:

$$\mathrm{F}^T_{tu} \ = \ (T^B, T^D, T^A)$$

Its rank and physical interpretation depend on the stated branch: the independent-d family and active conditional d = L interior have rank three, while completed closure on the latter branch has rank at most two. The temporal metric is open.

Contact and Realization

The event sequence is:

$$B \ \rightarrow \ @ \ \rightarrow \ D \ \rightarrow \ -@ \ \rightarrow \ A$$

@ actualizes an admissible pair contact. Each elementary pair interaction is one FTU; compatible contacts admit concurrent actualization as distinct events. C_1 controls entry, C_2 conditions the realized path, and C_3 determines what persists:

$$\Theta_{\text{con}} \ = \ \{C_1[@, \Phi],\ C_2[\Phi, \delta, \Sigma],\ C_3[\Phi, \Sigma]\}$$

Only D carries active xyz realization. At A, active depth ends and the completed x-y readout is retained in two endpoint record instances sharing one event identity. Both records reference one event-level source energy. Completed phase history accumulates B → (B,D) → (B,D,A).

PMM describes the same physical process from two complementary vantages:

Single-FTU vantage = the internal B → D → A passage of one shared event.

Multi-FTU vantage = the relations among completed and subsequent events.

Each FTU is a complete event and participates in the evolving network of other FTUs. These are two descriptions of one physical history.

Angular Temporal Recursion Principle (extension first-principle postulate). Completed A records, accessed through Φ in causal order, generate the potential B state of subsequent FTUs. Their angular relationships permit interference when the contributing channels are coherent. The $c^3 \to \Phi \to c^0$ connection runs *between* events; each individual FTU progresses $c^0 \to c^1 \to c^2 \to c^3$. Video recording and playback provide a causal consistency example, while coherent two-contribution interference supplies a conditional mathematical test. The native $\Phi \to$ B generation law and preservation of coherent relative phase await derivation.

PART I — CORE ARCHITECTURE

Compact First Principles

This section collects the core PMM definitions, representations, and physical readouts in one compact reference.

Foundational purpose. PMM provides an event architecture connecting Quantum Mechanics (QM), General Relativity (GR), and Information Theory (IT). Recovery of established physics and the depth program are secondary to the B $\to$ D $\to$ A event architecture.

Event hierarchy

$$T \;=\; \text{Time}$$

B = Before = Potential $\to$ Quantum Mechanics $\to$ Wave $\to$ Uncertainty $\to$ Pre-spatial

D = During = Realization $\to$ General Relativity $\to$ Particle $\to$ Partial Certainty $\to$ Spatial xyz

A = After = Transcription $\to$ Information Theory $\to$ Cube $\to$ Full Certainty $\to$ Post-spatial

$$T \;\approx\; B \circ D \circ A$$

Angular Temporal Recursion Principle (extension postulate) = causally prior A histories, collected in Φ, generate subsequent B wave possibilities through angular history-dependent combination; coherent channels can interfere.

Across-event grade bridge = $c^3 \to \Phi \to c^0$, always connecting distinct FTUs under causal order.

An individual FTU keeps its native $c^0 \to c^1 \to c^2 \to c^3$ progression.

Uncertainty $\to$ Partial Certainty $\to$ Full Certainty is PMM's qualitative event-state heuristic across B $\to$ D $\to$ A. It describes event status rather than complete microscopic knowledge.

$$@ \;=\; \text{Wave Function Collapse / Contact}$$

$$-@ \;=\; \text{Anti Contact}$$

Spatial geometry and contact window

$$x \;=\; \text{Height}$$

$$y \;=\; \text{Width}$$

$$z \;=\; \text{Depth}$$

Θ_{in} = opening factor — spatial manifestation grows from @ through D; opening-role name: Birth Sigmoid

Θ_{out} = closing factor — spatial manifestation closes toward $-@$; closing-role name: Record Sigmoid

Space Window = the bounded region where active space exists between the opening and the closing

Θ_{win} = Space Window = D-space manifestation/access window; it does not test whether underlying internal D geometry exists

$$q \;=\; \theta^D/\pi,\; 0 \;\leq\; q \;\leq\; 1$$

$$\Theta_{\text{in}} \;=\; \sin(\pi q/2);\; \Theta_{\text{out}} \;=\; \cos(\pi q/2)$$

$$\Theta_{\text{win}} \;=\; 2\Theta_{\text{in}}\Theta_{\text{out}} \;=\; \sin(\pi q)$$

Constants, fields, and constraints

$$k_B \;=\; \text{Boltzmann Constant}$$

c = Speed of Light Constant / realization crease / invariant still reference in D

$$m_p \;=\; \text{Planck Mass}$$

Φ = Accumulated record field

δ = Inter-participant distance (not the single-FTU depth z^D)

Σ = Closure spread (never a summation operator)

κ_{cl} = harmonic closure coefficient, distinct from the D-clock response κ

λ_G = gravitational source coupling; λ_{wave} denotes wavelength in the companion

s_i^A = active completed-record source assigned to a lineage node/dual cell, distinct from D progress q

α, β, ... = realized FTU event identifiers assigned after contact actualization

$\mathcal{P}_i$ = participant lineage derived from endpoint continuation under $\sim^L$

Q_e = projector representation of the C_1-admissible B subspace $\mathcal{H}_e^B[\Phi]$ for unordered pair e

d = length readout used in intrinsic T^D = d/c; from S^D/E^D = L/c, d = L follows if T^D is identified with the D action time S^D/E^D; this conditional consistency relation is not imposed outside that branch

$$t \;=\; \text{coordinate time used in } c^1 \;=\; d/t$$

τ = proper time; $d\tau^D$ = D proper-time increment

C_1 = Entry Constraint → What can enter the channel

C_2 = Path Constraint → What conditions the interaction

C_3 = Exit Constraint → What persists and is transcribed into Φ

Θ_{con} = Contact-information structure governing the conditions around @

$$\Theta_{\text{con}} \;=\; \{C_1[@,\; \Phi],\; C_2[\Phi,\; \delta,\; \Sigma],\; C_3[\Phi,\; \Sigma]\}$$

Pairwise realization

Elementary realized contact is pairwise.

Each elementary physically actualized unordered pair contact e@ = {i,j} = {j,i} is one FTU.

Before actualization, $C_1[\Phi]$ defines $\mathcal{H}_e^B[\Phi]$, the admissible B-state subspace for pair e.

$Q_e = \mathrm{Proj}(\mathcal{H}_e^B[\Phi])$ is the corresponding binary contact projector.

$$P_e@ \;=\; \langle \Psi_a^B d_m | Q_e | \Psi_a^B d_m \rangle \text{ and } P_e@ \;+\; P(\neg e@) \;=\; 1.$$

For a complete mutually exclusive block, $\sum P_e@ = 1$ is recovered inside that block.

Compatible commuting contacts admit concurrent actualization and shared participant lineages; each receives a distinct FTU event identifier after actualization.

$\sim^L$ = Lineage Continuity relation connecting endpoint occurrences of the same persisting carrier/history branch.

$\mathcal{P}_i$ = participant lineage generated by $\sim^L$ across incident FTUs.

The labels 1 and 2 order the two reciprocal internal D-vantage slots through c^1 and c^2; i and j identify participant lineages and do not own intrinsic c-grade identities.

$p_1|_e@ + p_2|_e@ = 1$ = conditional D-vantage seed weights within the selected pair once the internal reciprocal-channel decomposition is specified.

An external descriptive observer adds no FTU; a physical measurement/contact by that observer constitutes a new pairwise @ and a new FTU.

$$r_0 \;=\; \tfrac{1}{2} \ln(p_1|_e@ / p_2|_e@).$$

FTU architecture

$$\mathrm{F}_{tu} \;=\; B \circ D \circ A$$

The FTU physical c-order representation is:

$$B \;=\; (c^0)_{mp}$$

$$D \;=\; (c^1, c^2)c$$

$$A \;=\; (c^3)k_B$$

With its complete representation as:

$$\mathrm{F}_{tu}^c \;=\; ((c^0)_{mp}, (c^1, c^2)c, (c^3)k_B)$$

$$c^1 \;\mapsto\; D \text{ grade/slot 1 carrying } V_1 \;=\; (x_1^D, y_1^D)$$

$$c^2 \;\mapsto\; D \text{ grade/slot 2 carrying } V_2 \;=\; (x_2^D, y_2^D)$$

The correspondence assigns event grades to internal vantage slots. Participant identities are separate. The reciprocal D equations act through ρ_1, ρ_2, L, and r.

$$\rho_1 \;=\; \sqrt{x_1^D y_1^D}$$

$$\rho_2 \;=\; \sqrt{x_2^D y_2^D}$$

$$L \;=\; \sqrt{\rho_1 \rho_2} \;=\; \ell_E \theta^D \;=\; \text{common realized } D \text{ scale}$$

Scale = realized physical scale carried by D through L; expansion and contraction act through the existing common-scale and vantage-imbalance laws

z^D = relational depth generated across the two D vantages from the same common-scale geometry

FTU family glossary

$$\mathrm{F}_{tu} \;=\; \text{complete } B \;\to\; D \;\to\; A \text{ event}$$

$$\mathrm{F}^{c}_{tu} \;=\; c\text{-order representation}$$

$$\mathrm{F}^{M}_{tu} \;=\; \text{mass readout}$$

$$\mathrm{F}^{E}_{tu} \;=\; \text{energy readout}$$

$$\mathrm{F}^{T}_{tu} \;=\; \text{temporal state-space readout}$$

$$\mathrm{F}^{xy}_{tu} \;=\; \text{spatial readout}$$

$$\mathrm{F}^{H}_{tu} \;=\; \text{cumulative phase-history readout}$$

$\mathrm{F}^{\mathrm{Hxy}}_{tu}$ = combined phase-history / spatial readout

$$\mathrm{F}^{G}_{tu} \;=\; \text{gravity readout}$$

F^{s}_{tu} = FTU Action Cube — candidate coordinate product $S^B S^D S^A$, units $(\mathrm{J}\cdot\mathrm{s})^3$

$\mathrm{F}^{\mathrm{led}}_{tu}$ = FTU Action Ledger — additive phase action, units $\mathrm{J}\cdot\mathrm{s}$

$$\mathrm{F}^{sv}_{tu} \;=\; \text{phase-action vector}$$

$$\text{Mass} \;=\; \mathrm{F}^{M}_{tu} \;=\; (m_p,\; E^D/c^2,\; \hbar\omega^A/c^2)$$

$$\text{Energy} \;=\; \mathrm{F}^{E}_{tu} \;=\; (m_p c^2,\; E^D,\; \hbar\omega^A)$$

$$\text{Time} \;=\; \mathrm{F}^{T}_{tu} \;=\; (T^B, T^D, T^A) \;=\; ([\hbar/(m_p c^2)]\Delta\theta^B,\; d/c,\; \Delta\theta^A/\omega^A)$$

Temporal-rank result = the independent-d time map is generically rank three; on the conditional d = L branch the active D interior remains rank three while completed closure has rank at most two; no orthogonal metric is assumed

$$\text{Spatial} \;=\; \mathrm{F}^{xy}_{tu} \;=\; (\psi(x,y,z),\; \mathrm{xyz},\; xy)$$

Active spatial dimensionality = (0, 3, 0) across B, D, A

$$\text{History} \;=\; \mathrm{F}^{H}_{tu} \;=\; (B, (B,D), (B,D,A))$$

$$\text{Combined history / spatial} \;=\; \mathrm{F}^{\mathrm{Hxy}}_{tu} \;=\; (B, \psi(x,y,z)) \;\to\; ((B,D), \mathrm{xyz}) \;\to\; ((B,D,A), xy)$$

$$\text{Gravity} \;=\; \mathrm{F}^{G}_{tu} \;=\; \nabla\varphi;\; g \;=\; c^2\mathrm{F}^{G}_{tu}$$

Action Cube = $\mathrm{F}^{s}_{tu} = S^B \times S^D \times S^A$ — candidate, units $(\mathrm{J}\cdot\mathrm{s})^3$

$$\text{Action Ledger} \;=\; \mathrm{F}^{\text{led}}_{tu} \;=\; S^B \;+\; S^D \;+\; S^A$$

Physical c-order profile

$$c^0 \;=\; 1$$

$$c^1 \;=\; 299,792,458\ m/s \;=\; d/t$$

c^2 = 89,875,517,873,681,764 m^2/s^2 = E/m (massive rest-energy sector) = $E^2/|\mathbf{p}|^2$ (null sector)

$$c^3 \;=\; 26,944,002,417,373,989,539,335,912\ m^3/s^3 \;=\; \hbar G_N/\ell_p^2$$

The numerical identities are standard physics; PMM's physical claim is their ordered assignment to the $B \to D \to A$ event grades. Because the c grades carry different units, a graph of their raw SI numbers represents a specified unit choice. A unit-consistent logarithmic comparison uses $\log_{10}(X/X_0)$ with a stated positive reference X_0 of the same units as X.

D expansion / contraction readout:

$$X^D \;=\; q \;=\; \theta^D/\pi \;:\; 0 \;\to\; 1$$

$$C^D \;=\; \Theta_{\text{out}} \;=\; \cos(\theta^D/2) \;:\; 1 \;\to\; 0$$

The c-order ladder is the event-grade architecture of the FTU. The numerical escalation comes directly from exponentiation.

$$c^0 \;\leftrightarrow\; m_p$$

$$c^1 \;\leftrightarrow\; c$$

$$c^2 \;\leftrightarrow\; c$$

$c^3 \leftrightarrow$ A transcription grade

The crease roles are distinct:

$$m_p \;=\; \text{entry crease}$$

c = realization crease

k_B = transcription crease

D closes after the c^2 realization grades; c^3 is the A-side transcription grade. k_B converts the information readout into entropy units.

Core physical interpretations

Orientation: B-facing outwardness → A-facing inwardness

Gravity: gradient in D-phase time behavior

Light: "still" in the zero-proper-time sense

Entropy: A-phase persistent-record quantity

Space: emergent relational motion during D; realized relation supplies spatial separation and evolution of relation supplies motion

M_{op}: Motion Operations Principle — B carries potential relation/motion, D carries realized relation/motion, A records completed relation; acceleration/deceleration are changes in relational motion and directionality emerges from relational geometry

$$\text{Scale}: \text{ realized in } D \text{ through } L \;=\; \sqrt{\rho_1 \rho_2} \;=\; \ell_E \theta^D$$

Depth: distinct from ordinary height/width

Time: angular

$$\text{FTU}: \text{ full } B \;\rightarrow\; D \;\rightarrow\; A \text{ event}$$

FTU is geometry

Single-FTU vantage = Internal B → D → A event geometry

Multi-FTU vantage = Relational geometry among FTUs

An FTU has no isolated physical realization; physical geometry is relational among FTUs

Wave / Particle / Cube: B / D / A

Constants: phase creases

Native depth z^D is active in D and ends at transcription. The z argument in B is potential content, not active depth. After is accessed through retained information expressed in x/y representation.

Temporal inversion: active spatial realization follows $0 \rightarrow 3 \rightarrow 0$ while completed phase history accumulates B → (B,D) → (B,D,A); this describes phase history and active spatial realization. The dimensional-map interpretation is open.

Action provides a common measure across B, D, and A.

Two Vantages of PMM

One physical process admits two complementary descriptions. The single-FTU vantage follows a complete event. The multi-FTU vantage follows that event in relation to others.

Single FTU: creases + c-order + action

Multi-FTU: Φ + timing response + gravity

Within one FTU, the Ladder of Light orders the passage from possibility to realized geometry and completed record. The creases specify the phase relationships, and action measures the event across its phases.

The multi-FTU vantage describes how completed events influence later events. Persistent record accumulates in Φ, Φ conditions later D-phase timing, and relational differences in that timing produce the gravity response.

Completed record connects the two vantages. The c-order ladder defines a single FTU; gravity arises in the relational multi-FTU sector.

PART II — SINGLE-FTU VANTAGE

1. Fundamental Event

The Field Transition Unit is one complete passage from possibility through physical realization into completed record. It is the unit through which PMM describes the making and persistence of events.

$$\mathrm{F}_{tu} \;=\; B \circ D \circ A$$

with the phase boundaries:

$$B \;\to\; @ \;\to\; D \;\to\; -@ \;\to\; A$$

where:

- @ = contact / wave-function collapse / entry into realized D
- −@ = anti-contact / terminal D boundary / exit from realized D into A

The single-FTU vantage describes the internal geometry of one complete shared event. One elementary pairwise realization constitutes one FTU between an unordered participant pair e = {i,j}. The realized D event carries two ordered internal vantages V_1 and V_2 through the existing c^1/c^2 grades. The labels 1 and 2 belong to the event's D-vantage ordering with participant identities specified separately; exchanging V_1 and V_2 reverses the oriented imbalance while preserving the symmetric event geometry. An external observer (Player 3) describes the relationship; event count changes upon physical contact. If Player 3 physically measures or otherwise contacts a carrier of that completed event, Player 3 becomes a participant in a new pairwise contact; that new @ constitutes a second FTU with its own V_1/V_2 ordering and B → D → A history. If Player 4 independently performs another physical measurement/contact, that interaction constitutes a third FTU. When the Player 3 and Player 4 contacts are causally unordered, PMM represents the second and third FTUs as distinct concurrent events in one relational slice with one pairwise event identity for each contact. Both participant histories reference the same completed event through distinct endpoint A-record instances. The core retains one event-level E^D and therefore one event-level $E^A(\alpha)$; two endpoint records do not duplicate that energy. Any participant- or channel-dependent energy readout requires a separate stated bridge.

The multi-FTU vantage describes relational geometry among FTUs.

An FTU has no isolated physical realization. Physical world geometry is relational among FTUs.

PMM 3.7.6 develops the single-FTU vantage in Part II and the relational multi-FTU layer in Part III.

2. c-Order Event Ladder

The FTU uses four event grades:

```
B              D                    A
```

c^0 c^1 c^2 c^3

Interpretation:

$$c^0 \;=\; \text{pre-spatial grade}$$

c^1 = first D realization grade / first x-y vantage

c^2 = second D realization grade / second x-y vantage

c^3 = post-spatial transcription grade

PMM 3.7.6 uses the core D identification:

$$c^1 \;\mapsto\; V_1 \;=\; (x_1^D, y_1^D)$$

$$c^2 \;\mapsto\; V_2 \;=\; (x_2^D, y_2^D)$$

The arrow $\mapsto$ denotes a physical readout correspondence. c^1 and c^2 are realization grades, each carrying an x-y vantage on the same local D geometry. Spatial axes belong to those vantage readouts.

The event-grade assignment is:

$$B \;=\; (c^0)_{mp}$$

$$D \;=\; (c^1, c^2)c$$

$$A \;=\; (c^3)k_B$$

With its complete representation as:

$$\mathrm{F}^{c}_{tu} \;=\; ((c^0)_{mp}, (c^1, c^2)c, (c^3)k_B)$$

The native FTU ladder terminates at c^3 because the complete event contains:

1 B grade + 2 D grades + 1 A grade

The native FTU event grades are c^0 through c^3.

Physical c-Order Architecture — The Ladder of Light

PMM postulates the c-order ladder as the **physical event-grade architecture of the FTU**:

$$c^0 \;\rightarrow\; c^1 \;\rightarrow\; c^2 \;\rightarrow\; c^3$$

The numerical escalation across c^0, c^1, c^2, and c^3 comes directly from exponentiation. The Ladder of Light is the physical claim that the $B \rightarrow D \rightarrow A$ event architecture occupies this ordered structure in the correspondence:

$$\mathrm{F}^{c}_{tu} \;=\; ((c^0)_{mp}, (c^1, c^2)c, (c^3)k_B)$$

The tuple encodes three distinct crease roles:

$$m_p \;=\; \text{entry crease}$$

c = realization crease

k_B = transcription crease

The realization sequence is:

$$B \;/\; \text{entry} \;\rightarrow\; c^0$$

D_1 / realization $\rightarrow c^1$

D_2 / realization $\rightarrow c^2$

A / transcription $\rightarrow c^3$

The repeated c operation scales the event through the ordered grades. D occupies c^1 and c^2. At $-@$, D closes and the event enters c^3, the A-side transcription grade.

A is the terminal c^3 grade of the FTU architecture. k_B marks transcription into persistent record. Downstream c^4 coupling belongs to relations among completed records; its PMM derivation is OPEN.

This distinction is central:

$$\textbf{c}\text{-order} \;=\; \text{event-grade architecture}$$

k_B = transcription into a different category of quantity

$$\text{action} \;=\; \text{common ruler across } \mathbf{B},\ \mathbf{D},\ \text{and } \mathbf{A}$$

The c^1 and c^2 grades supply two realized D vantages:

$$c^1 \;\mapsto\; V_1 \;=\; (x_1^D, y_1^D)$$

$$c^2 \;\mapsto\; V_2 \;=\; (x_2^D, y_2^D)$$

Their relation supplies the D-side grade geometry. The variable r measures imbalance between the two vantage scales; relational depth and closure are built from their common geometry.

Logical status: the c-order correspondence is a PMM first-principle postulate for the single-FTU architecture. The exponentiation supplies the numerical scale; PMM supplies the physical assignment of those grades to entry, realization, D closure at $-@$, and A-side transcription.

The c-Order / Crease Map

The Ladder of Light is a **grade-and-crease map** of the phase-specific B, D, and A quantities.

Along the c-order axis:

$$c^0 \;\to\; c^1 \;\to\; c^2 \;\to\; c^3$$

The crease roles are:

$$m_p \;\to\; \text{entry}$$

c $\to$ realization across the D grades

$k_B \to$ transcription at the terminal $D \to A$ boundary

The event-grade climb therefore ends at c^3, the A transcription grade.

The $D \to A$ transition is:

c^2 completion $\to -@ \to c^3$ / A transcription $\to$ persistent record

At $-@$, realized spatial geometry is transcribed into the terminal c^3 persistent record.

c^3 Planck-Scale Remark and Gravitational-Crease Candidate

Standard-physics remark. At the terminal c-order grade, the Planck-length definition is:

$$\ell_p \;=\; \sqrt{\hbar G_N / c^3}$$

Equivalently:

$$c^3 \;=\; \hbar G_N / \ell_p^2$$

This is the standard Planck-length identity written in reverse. PMM assigns this standard relation to the terminal c^3 / A grade as an architectural correspondence.

PMM phase-history interpretation. The cumulative history readout is:

$$\mathrm{F}^{H}_{tu} \;=\; (B,(B,D),(B,D,A))$$

At the terminal c^3 / A transcription grade:

$$\mathrm{F}^{H}_{tu}|_{(c^3_{\cdot})} \;=\; (B,D,A)$$

Thus c^3 corresponds to the FTU grade at which complete BDA phase history is present for A-side transcription into persistent record. This supplies a native PMM interpretation of the terminal c^3 assignment; the numerical identity $c^3 = \hbar G_N/\ell_p^2$ continues to come from the standard Planck-length relation.

PMM closure-area readout. The current two-vantage geometry gives:

$$x^D y^D \;=\; L^2$$

and at completed D closure:

$$L_{\mathrm{cl}} \;=\; \pi\hbar c/E^D_{\mathrm{cl}}$$

Define the completed planar closure-area readout:

$$A_{\mathrm{cl}} \;=\; x^D_{\mathrm{cl}} y^D_{\mathrm{cl}} \;=\; L^2_{\mathrm{cl}}$$

therefore:

$$A_{\mathrm{cl}} \;=\; (\pi\hbar c/E^D_{\mathrm{cl}})^2$$

Define the gravitational-crease area from the same standard Planck relation:

$$A_G \;=\; \hbar G_N/c^3 \;=\; \ell_p^2$$

PMM then isolates a narrower candidate condition for the gravitational-crease limiting case. In ladder language, this is the condition that completed D closure reaches the Planck-area scale associated with the terminal c^3 correspondence:

$$A_{\mathrm{cl}} \;=\; A_G$$

If native C_3 persistence dynamics select this equality at that limiting boundary, then:

$$(\pi\hbar c/E^D_{\mathrm{cl}})^2 \;=\; \hbar G_N/c^3$$

which gives:

$$E^D_{\mathrm{cl}} \;=\; \pi\sqrt{\hbar c^5/G_N} \;=\; \pi m_p c^2$$

The factor π comes directly from the current closure postulate $\theta_{\mathrm{cl}} = \pi$ through $L_{\mathrm{cl}} = \pi\hbar c/E^D$.

and consequently:

$$L_{\mathrm{cl}} \;=\; \ell_p$$

$$A_{\mathrm{cl}} \;=\; \ell_p^2$$

The equality $A_{\mathrm{cl}} = A_G$ is a candidate condition for the gravitational-crease limiting boundary. Ordinary completed FTUs retain carrier-dependent L_{cl} and A_{cl}. This candidate belongs to the single-FTU closure/transcription sector. The multi-FTU gravity chain $\Phi \to \kappa \to \varphi \to$ g uses active-source extraction and clock response.

The terminal architectural sequence remains:

$c^2 \to -@ \to c^3$ / k_B transcription $\to$ A record

k_B remains the transcription crease beside terminal c^3; it is separate from the Planck identity and the closure-area candidate.

Action as the Ruler Across the Architecture

The c-order grades carry different dimensional character, and c^3 is the A-side transcription grade rather than another D spatial grade. PMM therefore uses action as the common measure across the B/D/A event:

$$\mathrm{F}_{tu}^{sv} \;=\; (S^B,\; S^D,\; S^A)$$

with:

$$[S^B] \;=\; [S^D] \;=\; [S^A] \;=\; \text{action}$$

The additive action ledger is:

$$\mathrm{F}_{tu}^{\mathrm{led}} \;=\; S^B \;+\; S^D \;+\; S^A$$

The same coordinates also define the candidate Action Cube $\mathrm{F}_{tu}^{s} = S^B S^D S^A$, units $(\mathrm{J}\cdot\mathrm{s})^3$ (§4.1.1). The ledger retains action units $\mathrm{J}\cdot\mathrm{s}$; the product is not substituted into additive action or energy relations.

Action provides the common ruler across the phase-distinct B, D, and A sectors.

The c-order plot rises through B and D to c^2, then reaches c^3 at A transcription. The $D \to A$ boundary changes the physical category from realized space to persistent record without adding a c^4 grade.

3. Creases

The phase creases are:

$$B \;\leftrightarrow\; m_p$$

$$D \;\leftrightarrow\; c$$

$$A \;\leftrightarrow\; k_B$$

Interpretation:

- $$m_p \;=\; \text{entry crease}$$
- c = realization crease
- k_B = transcription crease

Technical role:

$m_p \to$ anchors entry into the FTU architecture

c $\to$ scales D progression through the realization grades

$k_B \to$ transcribes A-side record multiplicity into entropy units

c is the realization-scaling crease; k_B is the transcription crease. The FTU ladder terminates at c^3.

3.1 Crease / Constant Scope Boundary

PMM accepts the fundamental constants used by the model, including $\hbar$, c, G_N, k_B, m_p, and associated standard Planck-scale identities, as physical inputs. Their origins and numerical values are accepted inputs.

Within PMM, these accepted constants act as physical creases / braces that constrain reconstruction of the B $\rightarrow$ D $\rightarrow$ A architecture and its measurable consequences. The model asks what event geometry, phase relations, and readouts are implied when these accepted physical quantities occupy their observed roles.

PMM derives event relations using these accepted constants.

4. Single-FTU Readouts

The primary FTU readouts use E^D for either carrier sector. The D mass slot E^D/c^2 is mass-equivalent energy; it equals rest mass m only in the massive rest-energy sector and is not a photon rest mass. Energy comparisons require a specified perspective/frame; differing readouts do not create additional shared events.

The primary FTU readouts are:

Mass $= \mathrm{F}^M_{tu} = (m_p,\ E^D/c^2,\ \hbar\omega^A/c^2)$

Energy $= \mathrm{F}^E_{tu} = (m_pc^2,\ E^D,\ \hbar\omega^A)$

Time $= \mathrm{F}^T_{tu} = ([\hbar/(m_pc^2)]\Delta\theta^B,\ d/c,\ \Delta\theta^A/\omega^A)$

Interpretation:

B time = projective potential span over the intrinsic conversion scale

$$D \text{ time } = \text{ realized traversal time}$$

A time = record span over the intrinsic transcription scale

Across the completed event, phase history accumulates B $\rightarrow$ (B,D) $\rightarrow$ (B,D,A). That cumulative history is distinct from the temporal state-space role of F^T_{tu}. The coordinates T^B, T^D, and T^A have branch-dependent independence; their inner products are open.

Derived Temporal Geometry of the Time Readout

EXISTING READOUT + DERIVED CONDITIONAL RESULT

The existing FTU time tuple supplies the physical temporal coordinates of one complete event:

$$\mathrm{F}^T_{tu} = (T^B, T^D, T^A)$$

with:

$$T^B = [\hbar/(m_pc^2)]\Delta\theta^B$$

$$T^D = d/c$$

$$T^A = \Delta\theta^A/\omega^A.$$

The temporal geometry is characterized by the rank of this time-readout map.

Derived temporal-rank result

DERIVED CONDITIONAL ON PA-B / PA-A — nondegenerate FTU family

Let:

$$\beta \;=\; \Delta\theta^{B}.$$

From the existing B/A probability relation:

$$P_{e}@ \;=\; \cos^{2}\beta$$

and:

$$\Delta\theta^{A} \;=\; -\mathrm{lnP}_{e}@ \;=\; -2\ln(\cos\beta) \;\equiv\; f(\beta).$$

Using $\omega^{A} = E^{D}/\hbar$, the physical time coordinates are:

$$T^{B} \;=\; [\hbar/(m_{p}c^{2})]\beta$$

$$T^{D} \;=\; d/c$$

$$T^{A} \;=\; [\hbar/E^{D}]f(\beta).$$

For the local physical input coordinates (β, d, E^{D}), define the Jacobian:

$$\mathrm{J}^{T} \;=\; \partial(T^{B}, T^{D}, T^{A})/\partial(\beta, d, E^{D}).$$

Its determinant is:

$$\det\ \mathbf{J^{T}} \;=\; -\hbar^{\mathbf{2}}\mathbf{f}(\boldsymbol{\beta})\ /\ [\mathbf{m_{p}c^{3}(E^{D})^{2}}].$$

For:

$$0 \;<\; \beta \;<\; \pi/2$$

and:

$$E^{D} \;>\; 0,$$

$$\textit{we have } f(\beta) \;>\; 0 \text{ and therefore } \det\ \mathrm{J}^{T} \;\neq\; 0. \text{ Hence :}$$

$$\mathrm{rank}\ \mathbf{J^{T}} \;=\; \mathbf{3}.$$

The physical FTU temporal map is therefore locally three-dimensional across the nondegenerate interior of the FTU family **on the parameterization in which d is an independent physical input**. At $P_{e}@ = 1$, $\beta = 0$ and $f(\beta) = 0$, so the map reaches a degenerate boundary. The opposite endpoint $P_{e}@ \to 0$ sends $f(\beta) \to \infty$ and is likewise a boundary of the finite-coordinate chart.

On the conditional D action-time branch d = L = $(\hbar c/E^{D})\theta^{D}$, use the active-D inputs $(\beta, \theta^{D}, E^{D})$. Then $T^{D} = \hbar\theta^{D}/E^{D}$ and the active interior remains generically rank three while θ^{D} is free. At completed closure $\theta^{D} = \pi$ is fixed, so the completed-event tuple depends only on (β, E^{D}) and its rank is at most two.

These rank statements depend on parameterization and phase status. Because $\Delta\theta^B$ and $\Delta\theta^A$ both depend on P_e@, their angular spans are coupled. Orthogonality, a Euclidean metric, and the temporal-cube interpretation are open metric questions.

The compact mathematics below expresses the remaining readouts in the PMM architecture.

4.1 Action as the Common B/D/A Measure

Action provides a common measure across B, D, and A. The c-order ladder assigns c^0 to B, c^1, c^2 to D, and c^3 to A transcription. Action provides the commensurable ruler across the full event.

Write the phase-action representation as:

$$\mathrm{F}_{tu}^{sv} \;=\; (S^B,\; S^D,\; S^A)$$

and the additive FTU action ledger as:

$$\mathrm{F}_{tu}^{\mathrm{led}} \;=\; S^B \;+\; S^D \;+\; S^A$$

where S^B, S^D, and S^A are the action contributions associated with the three sequential phases. The ledger measures the contributions; C_1, C_2, and C_3 govern entry, realization, and transcription.

PMM uses action as the common B/D/A ruler with variable phase-specific dimensionless spans:

$$S^B \;=\; \hbar\Delta\theta^B$$

$$S^A \;=\; \hbar\Delta\theta^A$$

and for D angular progression:

$$dS^D \;=\; \hbar\; d\theta^D$$

therefore:

$$S^D(\theta^D) \;=\; \hbar\theta^D$$

with $S^D(0) = 0$. Since the current single-FTU closure postulate is:

$$\theta_{\mathrm{cl}} \;=\; \pi$$

the completed D action readout is:

$$S^D, \mathrm{cl} \;=\; \pi\hbar$$

The existing D scale law also gives:

$$L \;=\; (\hbar c/E^D)\theta^D$$

so:

$$S^D/E^D \;=\; L/c$$

This is an exact algebraic consequence of the current D action and scale laws. If the existing D traversal-time readout $T^D = \mathrm{d/c}$ is required to be this same action time S^D/E^D, then:

$$d \;=\; L$$

Status: DERIVED CONDITIONAL CONSISTENCY RESULT. This relation applies to the D action-time branch. Proper-time identification additionally requires the timelike action bridge; distinct traversal-time branches keep d independent.

Energy from Angular Action. For a nonzero phase duration T and constant angular rate $\omega = \Delta\theta/T$, the angular-action relation $\mathrm{S} = \hbar\Delta\theta$ gives:

$$E \;=\; S/T \;=\; \hbar\omega$$

This provides an operational energy readout in joules from PMM's common action measure. For D, the relation applies on the identified action-time branch; B and A retain their phase-specific time and energy assignments.

The candidate B span is tied to the physical C_1-selected pair-contact channel. For the actualized pair e@ with pair-selection probability $P_e@$, define the projective contact span:

$$\Delta\theta^B \;=\; \arccos\sqrt{P_e@}$$

For rank-one Q_e, this is the ray-space angle between the normalized C_1-admissible B state and the selected pair-contact channel. For higher-rank Q_e it is the angle from the admissible B-state ray to the contact subspace $\mathcal{H}_e^B[\Phi]$. The normalized state $\Psi_a^B d_m$ is obtained by projecting Ψ^B with $\Pi_a d_m$ onto $\mathcal{H}_a^B d_m[\Phi]$ and renormalizing. The still-open physics is the $C_1[\Phi]$ law that determines the admissible subspaces themselves. The projective span is invariant under global phase and remains finite at the probability endpoints.

The A span is tied to the information carried by the actualized C_1-selected pair-contact channel. For the realized pair e@, define the single-event A information readout:

$$I^A \;=\; -\ln\; P_e@$$

and the associated effective A-side multiplicity:

$$\Omega_A \;=\; e^{I^A} \;=\; 1/P_e@$$

Therefore:

$$\Delta\theta^A \;=\; \mathcal{S}^A/k_B \;=\; \ln\;\Omega_A \;=\; I^A \;=\; -\ln\;P_e@$$

with $\mathcal{S}^A = k_B I^A$. For N equiprobable pair-contact alternatives, $P_e@ = 1/\mathrm{N}$ and $\Omega_A = \mathrm{N}$. The A span measures record information.

The completed elementary FTU action ledger is therefore:

$$\mathrm{F}_{tu}^{\mathrm{led}},\mathrm{cl} \;=\; \hbar(\Delta\theta^B \;+\; \pi \;+\; \Delta\theta^A)$$

The B span rule is a **candidate phase-action readout**; the A span rule is a **candidate physical information/action assignment**. They express phase spans in action units; c-order grades retain their phase-specific physical roles, and $C_1/C_2/C_3$ govern the transitions.

The D action is carrier-general because it is defined from angular progression. In the null sector, $d\tau^D = 0$ while $d\theta^D \neq 0$, so null D retains a defined angular-action readout without identifying action with proper time.

4.1.1 Vantage-Based Action Cube and Separate Action Ledger

Action Cube notation. F^{s}_{tu} denotes the candidate multiplicative Action Cube; $\mathrm{F}^{\mathrm{led}}_{tu}$ denotes the separate additive Action Ledger. F^{sv}_{tu} remains the coordinate tuple. The ledger carries action units. F^{L}_{tu} is reserved for FTU length and is not a ledger symbol.

Status: DEFINED coordinate operations / CANDIDATE geometric readout; native metric and predictive consequences OPEN. The Action Cube is an action-coordinate box, not an invariant spatial volume or an established temporal metric. Its edges need not be equal. Its three phase-based perspectives are distinct from the two internal D vantages V_1/V_2 and from the single-FTU/multi-FTU distinction.

Canonical definitions

`FTU ACTION COORDINATES`

$$\mathrm{F}^{sv}_{tu} \;=\; (S^B,\; S^D,\; S^A)$$

$$S^B \;=\; \hbar\Delta\theta^B$$

$$S^D(\theta^D) \;=\; \hbar\theta^D$$

$$S^A \;=\; \hbar\Delta\theta^A$$

`FTU ACTION CUBE (candidate action-coordinate product / volume)`

$$\mathrm{F}^{s}_{tu} \;=\; S^B \times S^D \times S^A$$

$$\mathrm{F}^{s}_{tu} \;=\; \hbar^3\Delta\theta^B\theta^D\Delta\theta^A$$

$$[\mathrm{F}^{s}_{tu}] \;=\; (\mathrm{J}\cdot\mathrm{s})^3$$

`FTU ACTION LEDGER (additive phase action)`

$$\mathrm{F}^{\mathrm{led}}_{tu} \;=\; S^B \;+\; S^D \;+\; S^A$$

$$\mathrm{F}^{\mathrm{led}}_{tu} \;=\; \hbar(\Delta\theta^B \;+\; \theta^D \;+\; \Delta\theta^A)$$

$$[\mathrm{F}^{\mathrm{led}}_{tu}] \;=\; \mathrm{J}\cdot\mathrm{s}$$

`CLOSURE`

$$\theta^D,\mathrm{cl} \;=\; \pi$$

$$S^D,\mathrm{cl} \;=\; \pi\hbar$$

$$\mathrm{F}^{s}_{tu},\mathrm{cl} \;=\; \pi\hbar^3\Delta\theta^B\Delta\theta^A$$

$$\mathrm{F}^{\mathrm{led}}_{tu},\mathrm{cl} \;=\; \hbar(\Delta\theta^B \;+\; \pi \;+\; \Delta\theta^A)$$

Meaning: Ledger = scalar sum of the phase-action readouts; Cube = candidate three-coordinate multiplicative readout. Multiplication is defined for an *action-coordinate box*. Interpretation as invariant physical geometric volume would require a metric / measure law, and is not asserted here.

Before During and After perspectives

The *same* event is represented at different stages. Candidate A-spans in Before and During are prospective; the actual A record is transcribed only at the $D \to A$ boundary.

BEFORE : $e \;\in\; \boldsymbol{\mathcal{E}}^B,\; 0 \;<\; P_e \;\leq\; 1$

$$\Delta\theta^B_e \;=\; \arccos(\sqrt{P_e})$$

$$\Delta\theta^A_e \;=\; -\ln(P_e)$$

$$S^B_e \;=\; \hbar\Delta\theta^B_e$$

$$S^D_e,\mathrm{cl} \;=\; \pi\hbar \qquad \text{(prospective closure value)}$$

$$S^A_e \;=\; \hbar\Delta\theta^A_e \qquad \text{(prospective } A \text{ value)}$$

$$\mathrm{F}^{s}_{tu},^{B}(e) \;=\; S^B_e \;\times\; S^D_e,\mathrm{cl} \;\times\; S^A_e$$

$$\mathrm{F}^{s}_{tu},^{B}(e) \;=\; -\pi\hbar^3\;\arccos(\sqrt{P_e})\;\ln(P_e)$$

$$\mathcal{V}^B \;=\; \{\mathrm{F}^{s}_{tu},^{B}(e) \mid e \;\in\; \boldsymbol{\mathcal{E}}^B,\; 0 \;<\; P_e \;\leq\; 1\}$$

`DURING: selected e@`

$\mathrm{F}^{s}_{tu}(\theta^D) \;=\; \hbar^3 \Delta\theta^B \theta^D \Delta\theta^A$

$0 \;\le\; \theta^D \;\le\; \pi$

`For selected B and prospective A spans held constant:`

$d\mathrm{F}^{s}_{tu} \;=\; S^B S^A dS^D$

$d\mathrm{F}^{s}_{tu}/d\theta^D \;=\; \hbar^3 \Delta\theta^B \Delta\theta^A$

$d\mathrm{F}^{\mathrm{led}}_{tu} \;=\; dS^D$

`AFTER: completed e@`

$\mathrm{F}^{s}_{tu}, \mathrm{cl} \;=\; -\pi\hbar^3 \;\arccos(\sqrt{P_e}@) \;\ln(P_e@)$

$\mathrm{F}^{\mathrm{led}}_{tu}, \mathrm{cl} \;=\; \hbar[\arccos(\sqrt{P_e}@) \;+\; \pi \;-\; \ln(P_e@)]$

Where $\boldsymbol{\mathcal{E}}^B$ denotes C_1-admissible pair contacts, P_e the candidate selection probability, and $\mathcal{V}^B$ the set of *prospective completed coordinate products*, rather than an assertion that a fully realized After volume physically exists in Before.

P_e is the existing C_1 binary pair-contact probability $P_e@$ before actualization; after selection, $P_e@$ continues to denote that original probability, not a probability reset to one by knowledge of the outcome. Distinct contacts can give the same product, so the map $\mathrm{e} \mapsto \mathrm{F}^{s}_{tu}{}^{,B}(e)$ retains identity when needed. After reconstruction concerns the supported action coordinates, not recovery of the full original wavefunction or microscopic trajectory. Endpoint accounting remains one event with two separate Afters and one event-level source energy.

For varying B or A spans the full product rule applies:

$$d\mathrm{F}^{s}_{tu} \;=\; S^D S^A \; dS^B \;+\; S^B S^A \; dS^D \;+\; S^B S^D \; dS^A.$$

$$d\mathrm{F}^{\mathrm{led}}_{tu} \;=\; dS^B \;+\; dS^D \;+\; dS^A.$$

The restricted differentials above hold only with B and prospective A spans fixed. Null angular progression is not replaced by null proper time. The two D c-grade vantages continue to share one S^D contribution in the ledger and one D coordinate in the cube.

Candidate interpretation and scope

$Q(P) \;=\; \Delta\theta^B \;\times\; I^A$

$I^A \;=\; -\ln\; P$

$\Delta\theta^B \;=\; \arccos(\sqrt{P})$

$Q(P) \;=\; -\arccos(\sqrt{P}) \;\ln\; P$

$\mathrm{F}^{s}_{tu}, \mathrm{cl} \;=\; \pi\hbar^3 Q(P)$

Candidate interpretation: information-weighted projective contact separation. Its proposed role is a B-to-A geometric/readout coupling, with physical consequences to be derived from PMM's $C_1/C_2/C_3$ architecture rather than adopted by analogy.

- At $\mathbf{P = 1}$, $\Delta\theta^B = I^A = 0$ and $\mathrm{F}^{s}_{tu}, \mathrm{cl} = 0$ although an event may complete. **Zero coordinate product does not mean zero closures.** Keep closure counting separate.
- As $\mathbf{P} \to 0^+$, Q(P) diverges; evaluate domain/normalization and admissible event probabilities.
- At completed closure, the cube and ledger are both functions of the same P, so the cube currently supplies a different nonlinear readout, **not an independent parameter**. During, the independent θ^D progression allows different pairs (P, θ^D) to be distinguished.
- B, D and A are phase-distinct and generally sequential. Orthogonality, metric independence, coordinate-to-physical volume conversion, and treatment of coupling are **open tests**. Do not use the product to claim an invariant spatial volume.

At P = 1, the cube vanishes but the completed ledger is $\pi\hbar$ and a closure may still occur. At P $\to 0^+$ the candidate product diverges; no divergent observable is asserted. A metric-dependent geometric volume requires an appropriate measure, for example $\sqrt{det(g)}\ dS^B dS^D dS^A$ for a positive-definite metric. The probability-coupled spans do not establish three freely independent physical axes. Closure count N is separate from the continuous coordinate product.

Canonical notation. The multiplicative Action Cube is a candidate geometric readout, distinct from the additive Action Ledger. B/A phase spans are variable and $\theta^D_{\rm cl} = \pi$; fixed B/A spans are not general identities.

FTU action glossary and reserved length notation

Symbol	Canonical name	Definition	Units	Status / usage
F^s_{tu}	**FTU Action Cube**	$S^B \times S^D \times S^A = \hbar^3 \Delta\theta^B \theta^D \Delta\theta^A$	$(\mathrm{J}\cdot\mathrm{s})^3$	Proposed multiplicative action-coordinate product; geometric/physical interpretation under test
$\mathrm{F}^{\rm led}_{tu}$	**FTU Action Ledger**	$S^B + S^D + S^A = \hbar(\Delta\theta^B + \theta^D + \Delta\theta^A)$	J · s	Existing additive action accounting, renamed from F^s_{tu}
F^{sv}_{tu}	**FTU Phase-Action Vector**	(S^B, S^D, S^A)	J · s per coordinate	Retained common action-coordinate representation
F^L_{tu}	**FTU Length**	Reserved for length; existing L depth/scale equations unchanged	m	Reserved for FTU length; not adopted here as a new length tuple and never used for the ledger

F^L_{tu} is reserved for length. The scale/depth relations are $\mathrm{L} = z^D = \sqrt{\rho_1\rho_2}$; a distinct length tuple is not defined here.

Symbol	Name	Definition	Units / role
Q(P)	Candidate information-weighted contact separation	$-\arccos(\sqrt{P})\ \ln \mathrm{P}$	Dimensionless; proposed cube closure factor
$\mathcal{V}^B$	Before prospective Action Cube set	$\{\mathrm{F}^{s,B}_{tu}(e) \mid \mathrm{e} \in \boldsymbol{\mathcal{E}}^B, 0 < P_e \le 1\}$	Set of $(\mathrm{J}\cdot\mathrm{s})^3$ values
$\boldsymbol{\mathcal{E}}^B$	C_1-admissible candidate pair contacts	Event-selection domain already defined by PMM	Set

Physical consequences for closure, persistence, energy or gravity require a separate native $C_1/C_2/C_3$ derivation. Defining Q(P) or the product does not supply that law or upgrade a recovery/prediction status.

4.2 c-Order Readout in the Common Action Unit

The c-order grades and action are distinct quantities. The common phase-action ruler maps the grades into action readouts:

c-order grade	PMM role	Action readout	Normalized $S/\hbar$	UOM
c^0	B	$S^B = \hbar\Delta\theta^B$	$\Delta\theta^B$	J · s
c^1	D vantage 1	$S^D = \hbar\theta^D$; at closure $\pi\hbar$	θ^D; at closure π	J · s
c^2	D vantage 2	$S^D = \hbar\theta^D$; at closure $\pi\hbar$	θ^D; at closure π	J · s
c^3	A	$S^A = \hbar\Delta\theta^A$	$\Delta\theta^A$	J · s

The c^1 and c^2 rows are two vantages of the same D phase. They therefore share the same S^D readout and are not counted as two separate D action contributions in the additive FTU ledger.

At completed single-FTU closure, the c-order action-readout profile is therefore:

$$\Delta\theta^B \;\rightarrow\; \pi \;\rightarrow\; \pi \;\rightarrow\; \Delta\theta^A$$

in units of $\hbar$. The B and A values are event-dependent rather than universal constants. This profile is a common-unit readout of the c-order architecture, not a replacement for the raw $c^0 \rightarrow c^1 \rightarrow c^2 \rightarrow c^3$ event-grade values.

5. B Phase — Potential

B is the pre-spatial possibility phase.

A B state is represented by:

$$\Psi^B \;=\; \sum_i \; a_i \; e^{i\theta_i} \; |i\rangle$$

with:

$$\sum_i \; |a_i|^2 \;=\; 1$$

C_1 controls entry into D.

For a two-component representation of the B state:

$$p_+ \;=\; |a_+|^2$$

$$p_- \;=\; |a_-|^2$$

$$p_+ \;+\; p_- \;=\; 1$$

These two-component representation weights are internal B-state weights. They are not the inter-participant pair-selection probabilities P_{ij}@ unless the representation has explicitly been chosen as that pair-contact basis.

5.1 Candidate C_1 Pair-Contact Selection

PMM separates **which unordered pair contact actualizes** from **how the two reciprocal D-vantage slots are seeded inside that realized pair**.

Let $\boldsymbol{\mathcal{E}}^B$ denote the C_1-admissible set of unordered candidate pair contacts at a relational update:

$$e \;=\; \{i,j\} \;=\; \{j,i\}.$$

No FTU event identifier is required before contact actualization.

For each candidate pair e, $C_1[\Phi]$ defines an admissible B-state subspace:

$$\mathcal{H}^B_e[\Phi] \;\subseteq\; \mathcal{H}^B.$$

Its sharp projector representation is:

$$Q_e \;=\; \mathrm{Proj}(\mathcal{H}^B_e[\Phi])$$

with:

$$Q_e^2 \;=\; Q_e$$

$$Q_e{}^\dagger \;=\; Q_e.$$

Define the total C_1-admissible B subspace as the span of the pair-admissible subspaces:

$$\mathcal{H}^B_a d_m[\Phi] \;=\; \mathrm{span}\{\mathcal{H}^B_e[\Phi] \;:\; e \;\in\; \boldsymbol{\mathcal{E}}^B\}.$$

Let:

$$\Pi_a d_m \;=\; \mathrm{Proj}(\mathcal{H}^B_a d_m[\Phi]).$$

For $\Pi_a d_m|\Psi^B\rangle \neq 0$, define the normalized C_1-admissible B state:

$$|\Psi^B_a d_m\rangle \;=\; \Pi_a d_m|\Psi^B\rangle \;/\; \sqrt{\langle\Psi^B|\Pi_a d_m|\Psi^B\rangle}.$$

This conditioning defines $\Psi^B_a d_m$ once C_1 has supplied the admissible subspaces; it does not derive the physical $C_1[\Phi]$ law itself.

For normalized C_1-admissible state $|\Psi^B_a d_m\rangle$, define the binary contact probability:

$$P_e@ \;=\; \langle\Psi^B_a d_m|Q_e|\Psi^B_a d_m\rangle$$

and:

$$P(\neg e@) \;=\; 1 \;-\; P_e@.$$

Therefore:

$$P_e@ \;+\; P(\neg e@) \;=\; 1.$$

Pair contacts have binary probabilities; complete exclusive blocks supply normalized alternative sets.

For mutually exclusive sharp alternatives:

$$Q_e Q_f \;=\; 0.$$

For a complete exclusive block $\boldsymbol{\mathcal{B}}$:

$$\sum_{e\in\boldsymbol{\mathcal{B}}} Q_e \;=\; I_{\boldsymbol{\mathcal{B}}}$$

and therefore:

$\sum_{e\in\boldsymbol{\mathcal{B}}} P_e@ = \langle \Psi^B_a d_m | I_{\boldsymbol{\mathcal{B}}} | \Psi^B_a d_m \rangle = 1$ when $\Psi^B_a d_m$ is supported in $\boldsymbol{\mathcal{B}}$; otherwise the probabilities must first be conditioned on that block.

A complete mutually exclusive contact block gives the normalized pair-selection rule.

For compatible concurrent sharp contacts:

$$[Q_e, Q_f] \;=\; 0$$

with:

$$Q_e Q_f \;\neq\; 0,$$

a joint yes-yes sector exists with:

$$P_e f@ \;=\; \langle \Psi^B_a d_m | Q_e Q_f | \Psi^B_a d_m \rangle.$$

If both contacts actualize, each unordered pair contact becomes its own FTU and receives its own event identifier after actualization. Compatible concurrent contacts admit shared participant lineages. For example, concurrent actualization of e = {i,j} and f = {i,k} is admitted when $[Q_e, Q_f] = 0$ and $Q_e Q_f \neq 0$. Lineage i then supplies one endpoint to each distinct FTU. Each FTU remains pairwise, and relational concurrency is defined by contact order. Noncommuting contacts $[Q_e, Q_f] \neq 0$ require a sequential or generalized measurement rule.

For a realized contact e@ with $P_e@ > 0$, define the normalized selected B state:

$$|\Psi^B_e\rangle \;=\; Q_e |\Psi^B_a d_m\rangle \;/\; \sqrt{P_e}@.$$

If the selected contact subspace admits two reciprocal internal channel projectors R_{e1} and R_{e2} satisfying:

$$R_{e1} R_{e2} \;=\; 0$$

$$R_{e1} \;+\; R_{e2} \;=\; Q_e,$$

then the D-vantage seed weights can be read from the selected state:

$$p_1|_e@ \;=\; \langle \Psi^B_e | R_{e1} | \Psi^B_e \rangle$$

$$p_2|_e@ \;=\; \langle \Psi^B_e | R_{e2} | \Psi^B_e \rangle$$

with:

$$p_1|_e@ \;+\; p_2|_e@ \;=\; 1.$$

The internal two-channel decomposition remains CANDIDATE pending native derivation. The labels 1 and 2 denote reciprocal D-vantage slots, not participant identities.

PMM then adopts the $B \to D$ **ratio-transfer postulate** at contact:

$$\lim_{\theta^D \to 0^+} \rho_1/\rho_2 \;=\; p_1|_e@ \;/\; p_2|_e@$$

Equivalently:

$$e^{2r_0} \;=\; p_1|_e@ \;/\; p_2|_e@$$

and therefore:

$$r_0 \;=\; \tfrac{1}{2}\ln(p_1|_e@/p_2|_e@).$$

The finite-r_0 reciprocal sector requires $0 < p_1|_e@,\, p_2|_e@ < 1$. Zero-weight endpoints are singular limits, not finite-r_0 D solutions.

Exchanging V_1 and V_2 sends $r_0 \to -r_0$ while leaving L, z^D, and Σ unchanged. Exchanging participant names alone has no physical effect.

Thus:

$$\text{full } B \text{ state } \Psi^B$$

$$\to\; C_1[\Phi] \text{ admissibility}$$

$$\to\; \text{pair-contact subspaces } \mathcal{H}_e^B[\Phi]$$

$$\to\; \text{binary contact projectors } Q_e$$

$$\to\; \text{actualized unordered pair } e@$$

$$\to\; \text{selected state } \Psi_e^B$$

$\to$ candidate reciprocal internal channel weights $p_1|_e@,\, p_2|_e@$

$$\to\; D\text{-vantage imbalance } r_0.$$

C_1 can therefore be read intuitively as:

what can enter the channel

Status: Pairwise Realization is an EXTENSION POSTULATE. The mapping $C_1[\Phi] \to \mathcal{H}_e^B[\Phi]$, the physical sharp-projector sufficiency of Q_e, and the internal reciprocal decomposition $R_{e1} + R_{e2} = Q_e$ remain CANDIDATE / OPEN physical structure. Q_e is the projector representation of the C_1-defined admissible subspace.

5.2 Candidate B Projective Span

Once C_1 physically selects the admissible pair-contact channels, the selected pair supplies a non-arbitrary endpoint for a B-side projective span. For the actualized pair e@:

$$\Delta\theta^B \;=\; \arccos\sqrt{P_e@}$$

with $P_e@$ given by the binary C_1 contact proposition of §5.1.

The associated B phase time and action readouts are:

$$T^B \;=\; [\hbar/E^B]\Delta\theta^B$$

$$S^B \;=\; E^B T^B \;=\; \hbar\Delta\theta^B$$

For the principal B energy $E^B = m_p c^2$:

$$T^B \;=\; [\hbar/(m_p c^2)]\Delta\theta^B$$

The intrinsic conversion scale is $T^B{*} = \hbar/E^B$; the physical B time is $T^B = T^B * \Delta\theta^B$.

Status: candidate projective B span. The contact basis is physically supplied by C_1 rather than chosen as an arbitrary reference.

Because $P_e@$ is the probability of the C_1-defined binary contact proposition for pair e, the associated A readout $I^A = -\ln P_e@$ is a **contact-conditioned surprisal**, not an intrinsic entropy of the pair alone. In a complete mutually exclusive block it also measures alternative-selection information. The same pair can therefore carry a different I^A when Φ changes its admissible contact subspace.

5.3 Candidate Reciprocal Closure Pairing

PMM tests a reciprocal B-side closure pairing as a candidate route to the Born rule. Inside a complete mutually exclusive C_1 contact block, the component probability readout is:

$$P_e \;=\; |a_e|^2 \;/\; \sum_k |a_k|^2$$

When the contact channels are represented by orthogonal projectors, this is the component form of the exclusive-block rule above. Outside a complete exclusive block, each pair-contact proposition retains its binary normalization rather than sharing one global $\sum P_e = 1$. The candidate remains subject to positivity, additivity where applicable, reciprocal symmetry, and pair-basis invariance.

6. D Phase — Realization

D is the spatial realization phase.

The two D realization grades provide two local spatial vantages:

$$c^1 \;\mapsto\; V_1 \;=\; (x_1^D(\theta^D),\; y_1^D(\theta^D))$$

$$c^2 \;\mapsto\; V_2 \;=\; (x_2^D(\theta^D),\; y_2^D(\theta^D))$$

These are two vantages on one local D geometry rather than four independent spatial dimensions.

Define the scalar scale of each vantage:

$$\rho_1 \;=\; \sqrt{x_1^D y_1^D}$$

$$\rho_2 \;=\; \sqrt{x_2^D y_2^D}$$

Define the common D scale and the c-grade imbalance coordinate:

$$L \;=\; \sqrt{\rho_1 \rho_2}$$

$$r \;=\; \tfrac{1}{2}\,\ln(\rho_1/\rho_2)$$

The logarithmic coordinates require positive vantage scales and positive planar lengths in the D interior. At $\theta^D = 0$, the vanishing scales and their ratios are interpreted by right-hand limits.

These invert exactly:

$$\rho_1 \;=\; L\, e^{r}$$

$$\rho_2 \;=\; L\, e^{-r}$$

Define the D energy length:

$$\ell_E \;=\; \hbar c/E$$

PMM adopts the **D scale postulate**:

$$L \;=\; \ell_E\, \theta^D$$

PMM separately adopts the minimal harmonic closure implementation:

$$r'' \;+\; \tfrac{1}{4} r \;=\; 0$$

Equivalently:

$$d^2/d\theta^{D2}\;\ln(\rho_1/\rho_2)$$

-

$$\tfrac{1}{4}\;\ln(\rho_1/\rho_2)$$

$$=$$

0

The scale postulate governs common realized scale.

The harmonic equation governs relational closure between the two c-grade vantages.

The four vantage scales x_1^D, y_1^D, x_2^D, y_2^D carry four independent modes: L, r, a and the vantage-shape disagreement

$$b \;=\; \tfrac{1}{4}\;\ln[(x_1^D/y_1^D)/(x_2^D/y_2^D)]$$

Together they invert exactly:

$$x_1^D \;=\; L\, e^{r+a+b}$$

$$y_1^D \;=\; L\; e^{r-a-b}$$

$$x_2^D \;=\; L\; e^{-r+a-b}$$

$$y_2^D \;=\; L\; e^{-r-a+b}$$

PMM imposes no vantage coherence during D. Planar-shape disagreement b $\neq 0$ is admitted during active D. Agreement is required only at the terminal vantage lock at $-@$ (§10, P3).

Define the common planar readouts:

$$x^D \;=\; \sqrt{x_1^D x_2^D}$$

$$y^D \;=\; \sqrt{y_1^D y_2^D}$$

and the planar-shape coordinate:

$$a \;=\; \tfrac{1}{2}\;\ln(x^D/y^D)$$

Then:

$$x^D \;=\; L\; e^a$$

$$y^D \;=\; L\; e^{-a}$$

The variables therefore separate different geometric roles:

L = common scale (even under $V_1 \leftrightarrow V_2$ and under $x \leftrightarrow y$)

r = imbalance between the c^1 and c^2 vantages (odd under $V_1 \leftrightarrow V_2$, even under $x \leftrightarrow y$)

a = average x-y shape within their common spatial plane (even under $V_1 \leftrightarrow V_2$, odd under $x \leftrightarrow y$)

b = disagreement between the vantages about x-y shape (odd under both exchanges)

The modes r and b cancel identically from $x^D = \sqrt{x_1^D x_2^D}$ and $y^D = \sqrt{y_1^D y_2^D}$, so $x^D = Le^a$, $y^D = Le^{-a}$ and $x^D y^D = L^2$ hold for all admitted r and b.

This separation prevents c-order grade from being identified with spatial axis.

7. Depth

Depth is generated through the relationship between the two During vantages. It has its own construction: the common relational scale of their realized spatial readouts. Depth therefore enters as a consequence of D geometry, rather than as an independently supplied third axis.

From:

$$x^D \;=\; \sqrt{x_1^D x_2^D}$$

$$y^D \;=\; \sqrt{y_1^D y_2^D}$$

and:

$$\rho_1 \;=\; \sqrt{x_1^D y_1^D}$$

$$\rho_2 \;=\; \sqrt{x_2^D y_2^D}$$

we have:

$$x^D y^D \;=\; \rho_1 \rho_2 \;=\; L^2$$

Therefore define the internal/local depth readout:

$$z^D \;=\; L \;=\; \sqrt{x^D y^D}$$

Depth is therefore the common relational scale generated across the two D vantages. Neither c^1 nor c^2 is identified with a single spatial axis.

7.1 Two-Vantage Symmetric / Antisymmetric Readout

From the existing D-vantage relations:

$$\rho_1 \;=\; Le^r$$

$$\rho_2 \;=\; Le^{-r}$$

define the symmetric and antisymmetric vantage readouts:

$$\rho_+^D \;=\; \tfrac{1}{2}(\rho_1 \;+\; \rho_2)$$

$$\rho_-^D \;=\; \tfrac{1}{2}(\rho_1 \;-\; \rho_2)$$

Therefore:

$$\rho_+^D \;=\; L \;\cosh\; r$$

$$\rho_-^D \;=\; L \;\sinh\; r$$

and:

$$(\rho_+^D)^2 \;-\; (\rho_-^D)^2 \;=\; L^2 \;=\; (z^D)^2$$

Under the existing pure c-grade rebalance:

$$\rho_1 \;\to\; \mu\rho_1$$

$$\rho_2 \;\to\; \rho_2/\mu$$

the common scale remains invariant while:

$$r \;\to\; r \;+\; \ln\,\mu$$

Thus $z^D = \mathrm{L}$ is invariant under reciprocal vantage rebalance, while r and ρ_-^D encode the relative imbalance between the two D vantages.

Status: DERIVED READOUT. These relations follow directly from the existing two-vantage geometry and introduce no time, velocity, inertial-frame, or spacetime interpretation.

7.2 Candidate Physical Instantiation of V_1 / V_2

The current two-vantage geometry admits a candidate physical interpretation in which V_1 and V_2 are reciprocal null/light-channel vantages of the same realized D interaction. Each channel carries an opposed propagation orientation together with a measurable phase/frequency or equivalent scale readout.

$V_1 \leftrightarrow +$ null / light channel

$V_2 \leftrightarrow -$ null / light channel

ρ_1, $\rho_2 \leftrightarrow$ reciprocal channel-scale readouts

$\mathrm{r} = \frac{1}{2}\ln(\rho_1/\rho_2) \leftrightarrow$ relative channel imbalance

$$z^D \;=\; \sqrt{\rho_1\rho_2} \;\leftrightarrow\; \text{common } D \text{ scale}$$

This candidate is physically motivated by counter-propagating light-channel configurations whose optical or laser realizations produce normalized reciprocal channel factors. Experimental correspondence and any Lorentz / special-relativity comparison belong to separate recovery and testing work rather than the native first-principles construction.

Status: CANDIDATE PHYSICAL INTERPRETATION. The universal channel identification requires a native degree-of-freedom derivation and independent experimental tests.

7.3 FTU Spatial Readout

Across the full $B \;\to\; D \;\to\; A$ event :

$$\mathrm{F}_{tu}^{xy} \;=\; (\psi(x,y,z),\ \mathrm{xyz},\ xy)$$

B: $\psi(x,y,z)$ = potential x/y/z content before spatial realization; x, y, and z are not active B coordinates

D: xyz = active spatial geometry; z^D is relational depth generated across the two D vantages

A: xy = retained post-spatial representation after active depth z^D ends

The z argument in $\psi(x,y,z)$ is potential content for possible D realization; it is not native depth itself. Active spatial dimensionality therefore follows $0 \to 3 \to 0$ across B $\to$ D $\to$ A, while representational content follows $\psi(x,y,z) \to$ xyz $\to$ xy.

Define the cumulative FTU phase-history readout:

$$\mathrm{F}_{tu}^{H} \;=\; (B,(B,D),(B,D,A))$$

and combine it with the spatial readout as:

$$\mathrm{F}_{tu}^{\mathrm{Hxy}} \;=\; (B,\psi(x,y,z)) \;\to\; ((B,D),\mathrm{xyz}) \;\to\; ((B,D,A),xy)$$

F_{tu}^{H} records completed phase accumulation while F_{tu}^{xy} records spatial content at each B/D/A stage. $\mathrm{F}_{tu}^{\mathrm{Hxy}}$ displays both readouts together; Phase history and active spatial dimensionality are distinct readouts. Temporal inversion describes their opposing pattern; the 3+1 $\to$ 1+3 dimensional interpretation is open.

The B entry uses $\psi(x, y, z)$ as the spatial-potential form of the B state; the canonical state notation remains Ψ^B.

Under a pure c-grade rebalance:

$$\rho_1 \ \rightarrow \ \mu\rho_1$$

$$\rho_2 \ \rightarrow \ \rho_2/\mu$$

L and z^D remain invariant.

At closure:

$$r \ \rightarrow \ 0$$

so:

$$\rho_1 \ = \ \rho_2 \ = \ L$$

The two c-grade vantages reach scale agreement and, under the vantage lock, shape agreement (b = 0). Closure locks the vantages; a independently specifies the completed planar shape.

7.4 Depth versus Distance

PMM keeps two relational quantities apart.

Depth (z^D) is internal to one FTU. It is generated between the two x-y vantages V_1 and V_2 of that FTU:

$$z^D \ = \ L \ = \ \sqrt{\rho_1\rho_2}$$

It is native to D and ends at the D → A boundary. One FTU has depth. It has no distance.

Distance (δ_{ij}) is between two participants, that is, between two realized three-dimensional events, each of which already carries its own x, y and depth. It belongs to the multi-FTU sector and is an independent primitive there (§21). No single FTU's vantages generate it.

Length readout (d) is the single-FTU traversal length that enters T^D = d/c. It is neither depth nor distance. The current D action and scale laws give S^D/E^D = L/c; therefore d = L follows on the conditional branch where T^D = d/c is identified with that same D action time. Outside that branch, d remains a distinct traversal-length readout. The candidate edge bridge $d\alpha \leftrightarrow \delta_{ij}$ (§34.10) relates it to distance.

Quantity	Where	Between	Status
Depth $z^D = \mathrm{L}$	inside one FTU, D only	the two x-y vantages	derived from the vantage construction
Length readout d	inside one FTU	the D traversal	input to $T^D = \mathrm{d/c}$; $\mathrm{d} = \mathrm{L}$ on the conditional D action-time branch
Distance δ_{ij}	multi-FTU network	two participants	independent primitive

Axes. Depth has an axis, the z^D direction of one FTU. Distance has no primitive direction. Direction is reconstructed only after the distance network is embedded (§21). The separation direction between two participants and the depth axis of each participant are therefore different objects, and the core does not identify them.

The direct bridge between depth and distance remains an open derivation target.

8. Closure Spread

Define the closure spread across the two D-vantage scales:

$$\Sigma \;=\; \rho_1 \;+\; \rho_2 \;-\; 2z^D$$

Using:

$$\rho_1 \;=\; Le^{r}$$

$$\rho_2 \;=\; Le^{-r}$$

$$z^D \;=\; L$$

gives:

$$\Sigma \;=\; 2L(\cosh\; r \;-\; 1)$$

or:

$$\Sigma \;=\; 4L\;\sinh^2(r/2)$$

Therefore:

$$\Sigma \;\geq\; 0$$

and, for $\mathrm{L} > 0$:

$$\Sigma \;=\; 0$$

$$\Leftrightarrow$$

$$r \;=\; 0$$

$$\Leftrightarrow$$

$$\rho_1 \;=\; \rho_2 \;=\; z^D \;=\; L$$

At entry, L = 0 gives $\Sigma = 0$ in the right-hand limit even when $r_0 \neq 0$; this entry zero is not terminal closure.

Σ measures the scale imbalance of the two active D vantages; a measures x-y planar anisotropy.

9. C_2 — Native D Evolution

C_2 governs the active passage through During. As angular progress advances, the shared scale expands while imbalance between the two vantages contracts. The event approaches closure by evolving its own relational geometry.

The primitive D evolution is written on the two vantage scales:

$$L \;=\; \sqrt{\rho_1 \rho_2} \;=\; \ell_E \, \theta^D$$

and:

$$d^2/d\theta^{D2} \;\ln(\rho_1/\rho_2)$$

-

$$\tfrac{1}{4} \;\ln(\rho_1/\rho_2)$$

$$=$$

0

For compact analysis:

$$r \;=\; \tfrac{1}{2} \;\ln(\rho_1/\rho_2)$$

so the same system becomes:

$$L \;=\; \ell_E \, \theta^D$$

and:

$$r'' \;+\; \tfrac{1}{4} r \;=\; 0$$

PMM 3.7.6 takes the single-FTU closure interval as the explicit closure postulate:

$$\theta_{\mathrm{cl}} \;=\; \pi$$

P5 is stated as the **minimal linear conservative closure ansatz**. Require the closure law to be linear, autonomous, second-order, undriven, undamped, and time-reversal symmetric. The minimal equation is then:

$$r'' \;+\; \kappa_{\mathrm{cl}} \, r \;=\; 0$$

For $r_0 \neq 0$, with the entry-symmetry condition $r'(0) = 0$ and the first zero required at $\theta^D = \pi$, the coefficient is conditionally derived:

$$\kappa_{\mathrm{cl}} \;=\; \tfrac{1}{4}$$

PMM adopts the harmonic class, and the numerical coefficient follows inside that class.

With the **entry-symmetry condition**:

$$r(0) \;=\; r_0$$

$$r'(0) \;=\; 0$$

where PMM adopts $r'(0) = 0$ as the minimal time-symmetric entry condition,

the solution is:

$$r(\theta^D) \;=\; r_0 \;\cos(\theta^D/2)$$

Expansion–Contraction Readout

The existing D equations supply two derived readouts across the same angular progression:

$$L \;=\; \ell_E \theta^D$$

$$r \;=\; r_0 \;\cos(\theta^D/2)$$

The common realized scale L expands with θ^D, while the magnitude of the c-grade vantage imbalance contracts toward zero throughout $0 < \theta^D < \pi$ when $r_0 \neq 0$. Define the normalized readouts:

$$X^D \;=\; q \;=\; \theta^D/\pi$$

$$C^D \;=\; \Theta_{\text{out}} \;=\; \cos(\theta^D/2)$$

Across the single-FTU D interval $0 \;\leq\; \theta^D \;\leq\; \pi$:

$$X^D \;:\; 0 \;\rightarrow\; 1$$

$$C^D \;:\; 1 \;\rightarrow\; 0$$

Expansion and contraction are therefore two readouts of the same D evolution. They summarize the existing D scale law and harmonic closure solution; they add no postulate beyond the stated scale and closure assumptions. The additional identification $C^D = \Theta_{\text{out}}$ matches the P5 contraction to the P6 closing factor because both use $\theta_{\text{cl}} = \pi$ and the selected half-angle forms; it is not a derivation of P6 from P5.

So D simultaneously carries:

outwardness = common realization scale growth

$$\text{inwardness} \;=\; c\text{-grade vantage closure}$$

Interpretation:

D is outward realization plus inward completion.

$$C_2 \text{ is therefore :}$$

what happens in the channel

Candidate D Half-Angle Carrier

The harmonic closure equation admits an exact two-component first-order representation. Define the unnormalized D half-angle carrier:

$$\tilde{\Xi}^D \;=\; (r,\; -2r')^T$$

Using:

$$r'' \;+\; \tfrac{1}{4}r \;=\; 0$$

gives:

$$d\tilde{\Xi}^D/d\theta^D \;=\; \tfrac{1}{2}\;[[0,-1],[1,0]]\;\tilde{\Xi}^D$$

For the entry-symmetry solution:

$$r(\theta^D) \;=\; r_0\;\cos(\theta^D/2)$$

therefore:

$$\tilde{\Xi}^D(\theta^D) \;=\; r_0(\cos(\theta^D/2),\;\sin(\theta^D/2))^T$$

This form preserves both the magnitude and sign of r_0. Since:

$$r_0 \;=\; \tfrac{1}{2}\ln(p_1|_e@/p_2|_e@)$$

the carrier retains which D vantage dominates at entry.

For the degenerate equal-weight case:

$$p_1|_e@ \;=\; p_2|_e@ \;\Rightarrow\; r_0 \;=\; 0 \;\Rightarrow\; \tilde{\Xi}^D \;=\; (0,0)^T$$

so no oriented half-angle carrier is assigned.

At entry:

$$\tilde{\Xi}^D(0) \;=\; (r_0,0)^T$$

and at canonical single-FTU closure:

$$\tilde{\Xi}^D(\pi) \;=\; (0,r_0)^T$$

Thus one completed D interval carries the nondegenerate carrier through a quarter turn in its internal two-component phase plane. This statement concerns the carrier plane generated by the D harmonic law. θ^D is PMM angular progress, and no identification with a spatial rotation angle is made here.

The mathematical continuation obeys:

$$\tilde{\Xi}^D(2\pi) \;=\; -\tilde{\Xi}^D(0)$$

$$\tilde{\Xi}^D(4\pi) \;=\; \tilde{\Xi}^D(0)$$

so the half-frequency oscillator has 4π periodicity. This periodicity follows from the half-angle harmonic form. Spin-$\frac{1}{2}$ interpretation requires rotational generators and an orientation bridge.

The first-order generator is a real planar rotation generator. Algebraically:

$$\tfrac{1}{2}\ [[0,-1],[1,0]]\ =\ -(i/2)\sigma_y$$

but the native carrier remains a real two-vector with one generated rotation axis. PMM therefore has an SO(2)-type half-angle carrier at this stage. A full SU(2) interpretation requires additional noncommuting generators and a physical bridge from θ^D progression to spatial rotational orientation.

The existing c-grade rebalance remains:

$$\rho_1\ \rightarrow\ \mu\rho_1$$

$$\rho_2\ \rightarrow\ \rho_2/\mu$$

which is a real reciprocal squeeze and supplies no second unitary rotation generator by itself.

Notation boundary: Σ remains the canonical Closure Spread. The half-angle carrier uses $\tilde{\Xi}^D$.

Status: the two-component first-order form, closure quarter turn, and 4π mathematical periodicity follow from P5. A physical spinor interpretation, full SU(2), spatial-rotation identification, and cross-FTU orientation persistence remain CANDIDATE / OPEN.

10. Angular Closure

The single-FTU closure postulate is:

$$\theta_{\mathrm{cl}}\ =\ \pi$$

Using the minimal harmonic closure law:

$$r(\theta^D)\ =\ r_0\ \cos(\theta^D/2)$$

gives:

$$r(\pi)\ =\ 0$$

Therefore at closure:

$$r\ =\ 0$$

$$\Sigma\ =\ 0$$

$$\rho_1\ =\ \rho_2\ =\ z^D\ =\ L_{\mathrm{cl}}$$

with:

$$L_{\mathrm{cl}}\ =\ \pi\hbar c/E^D$$

The common planar readouts remain:

$$x^D\ =\ L_{\mathrm{cl}}\ e^a$$

$$y^D \;=\; L_{\mathrm{cl}}\, e^{-a}$$

so closure requires the two c-grade vantages to agree (r = 0 and, by the vantage lock, b = 0) rather than $x^D = y^D$.

Anti-contact is:

$$-@ \;\Leftrightarrow\; \theta^D \;=\; \pi \;\Rightarrow\; r \;=\; 0,\; b \;=\; 0 \text{ and } \Sigma \;=\; 0$$

This is the terminal vantage lock (P3). At $-@$ the two vantages agree in scale (r = 0, $\rho_1 = \rho_2$) and in planar shape (b = 0), so $V_1 = V_2$ and the shared D geometry has one completed x-y readout. P3 closes D geometry; it does not identify the two physical endpoint occurrences with one A-record instance. During D the vantages need not agree. C_3 transcribes the common completed shape a_{cl}, after both disagreement modes have closed, into one persistent record instance at each lineage endpoint.

CANDIDATE: b follows the same minimal closure class as r, $b(\theta^D) = b_0 \cos(\theta^D/2)$, which gives b = 0 at $\theta^D = \pi$ without separate imposition. If b follows another law, the lock b = 0 at $\theta^D = \pi$ remains the stated condition.

For finite $r_0 \neq 0$ and $0 < \theta^D \leq \pi$, r = 0 and $\Sigma = 0$ occur only at $\theta^D = \pi$. For balanced entry $r_0 = 0$, both vanish throughout D; $-@$ is still fixed by $\theta^D = \pi$.

The fixed π closure is a single-FTU postulate. The harmonic form is the minimal linear conservative closure ansatz; once that class is adopted, the coefficient $\frac{1}{4}$ follows from requiring the first closure at π.

11. Space Window

Normalize D progression:

$$q \;=\; \theta^D \;/\; \theta_{\mathrm{cl}}$$

with:

$$0 \;\leq\; q \;\leq\; 1$$

Phase-progress notation: u denotes generic monotonic B → D → A ordering. D uses angular progress θ^D and normalized progress q = θ^D/π, with $\theta_{\mathrm{cl}} = \pi$. Window notation $\Theta_{\mathrm{in}}(u)$, $\Theta_{\mathrm{out}}(u)$, and $\Theta_{\mathrm{win}}(u)$ uses u as the ordering parameter.

Window terminology: Birth Sigmoid and **Record Sigmoid** name the opening and closing roles of Θ_{in} and Θ_{out}. Their mathematical forms are the angular half-window factors below.

PMM adopts the **lowest-mode symmetric space-window ansatz**:

$$\Theta_{\mathrm{in}} \;=\; \sin(\pi q/2)$$

$$\Theta_{\mathrm{out}} \;=\; \cos(\pi q/2)$$

The window is required to satisfy:

$$\Theta_{\mathrm{win}}(0) = 0$$

$$\Theta_{\mathrm{win}}(1) = 0$$

$$\Theta_{\mathrm{win}}(q) > 0 \text{ for } 0 < q < 1$$

$$\Theta_{\text{win}}(q) = \Theta_{\text{win}}(1-q)$$

$$\max\ \Theta_{\text{win}}\ =\ 1$$

Those conditions admit multiple functions. PMM selects the smooth positive fundamental mode with no internal nodes or extra oscillations. Within that class:

$$\Theta''_{\text{win}}\ +\ \pi^2\Theta_{\text{win}}\ =\ 0$$

with the endpoint conditions above uniquely selects:

$$\Theta_{\text{win}}\ =\ \sin(\pi q)$$

PMM chooses the half-window factors as the symmetric factorization:

$$\Theta_{\text{in}}\ =\ \sin(\pi q/2)$$

$$\Theta_{\text{out}}\ =\ \cos(\pi q/2)$$

Define this FTU's internal/local spatial-window readout:

$$\Theta_{\text{win}}\ =\ 2\ \Theta_{\text{in}}\ \Theta_{\text{out}}$$

therefore:

$$\Theta_{\text{win}}\ =\ \sin(\pi q)$$

The active spatial readout is:

(x,y,z)

$$=$$

$$\Theta_{\text{win}}\ (x^D, y^D, z^D)$$

So:

@:

$$q\ =\ 0$$

$$\Theta_{\text{win}}\ =\ 0$$

D:

$$0\ <\ q\ <\ 1$$

$$\Theta_{\text{win}}\ >\ 0$$

$$-@:$$

$$q \;=\; 1$$

$$\Theta_{\text{win}} \;=\; 0$$

The two D vantages reach relational completion as the active spatial window closes. Θ_{win} is a D-space manifestation/access window; it is not a measure of whether the underlying internal D geometry exists.

This separates:

internal D geometry

from:

active spatial manifestation

12. Outwardness and Inwardness

PMM phase orientation is:

$$B \;=\; \text{outward possibility}$$

D = outward realization + inward closure

$$A \;=\; \text{inward persistence}$$

Technical form:

$$dL/d\theta^D \;>\; 0$$

represents outward realization.

$$d|r|/d\theta^D \;<\; 0$$

throughout $0 < \theta^D < \pi$ when $r_0 \neq 0$ represents inward completion.

The two tendencies coexist during D.

13. Massive Sector

For a massive D realization:

$$E^D \;=\; mc^2$$

therefore:

$$\ell_E \;=\; \hbar c/E^D$$

$$=\; \hbar/(mc)$$

and:

$$L \;=\; (\hbar/(mc))\;\theta^D$$

The intrinsic D traversal-time readout is:

$$T^D \;=\; d/c$$

From the existing D action and scale laws:

$$S^D/E^D \;=\; L/c.$$

Therefore, if T^D is identified with the D action time S^D/E^D, then:

$$d \;=\; L.$$

Massive realization accumulates proper time. The further identification of this D action time with proper time uses the standard timelike action bridge:

$$dS^D \;=\; E^D d\tau.$$

With $dS^D = \hbar d\theta^D$ and $E^D = mc^2$, this gives dL = c $d\tau$ and, from D entry, L = $c\tau$. On that massive/timelike branch:

$$d \;=\; L \;=\; c\tau$$

$$T^D \;=\; \tau.$$

This proper-time identification is conditional on the timelike action bridge and does not extend to the null sector. The coordinate/relational comparison time t used to compare clock rates remains distinct from T^D.

14. Null Sector

The null sector has its own native definition.

For a null carrier:

$$E\gamma \;=\; \hbar\omega$$

For the trajectory and derivatives below, $E\gamma$ and ω are constant over the D interval in the specified coordinate-time frame.

therefore:

$$\ell_E \;=\; \hbar c/E\gamma$$

$$=\; c/\omega$$

and:

$$L\gamma \;=\; (c/\omega)\;\theta^D$$

with:

$$d\theta^D \;=\; \omega\; dt$$

so:

$$dL\gamma \;=\; c\;dt$$

therefore:

$$dL\gamma/dt \;=\; c$$

while:

$$d\tau^{D} \;=\; 0$$

Light therefore progresses through D by phase progression while accumulating zero proper time.

This is the PMM meaning of:

light as stillness

14.1 Null D-Propagation Interpretation

For an undisturbed null carrier, combine:

$$dL\gamma/dt \;=\; c \;>\; 0$$

with:

$$d\tau^{D} \;=\; 0$$

and the existing PMM definition:

$dL/d\theta^{D} > 0$ = outward realization.

Therefore the null carrier accumulates zero proper time while its realized D geometry progresses outward at c. In PMM's human-facing language: **the photon is still; During propagates.** Exterior photon motion is read as outward progression of the D realization associated with the null event.

Status: DERIVED INTERPRETATION. This combines existing null-sector and outward-realization relations; it introduces no photon rest frame, new dynamics, or additional primitive.

The massive and null sectors share the formal scale definitions below, not one universal physical proper-length interpretation. Under the candidate massive/timelike bridge L $\leftrightarrow$ $c\tau$, L is a Lorentz-invariant proper scale. For null propagation L = $c\Delta t$ and $L_{\mathrm{cl}} = \lambda_{\mathrm{wave}}/2$ are frame-dependent length readouts while $\Delta\tau = 0$. Algebraic invariance of $\sqrt{\rho_1\rho_2}$ under reciprocal vantage rebalance is a separate property and does not make a photon's propagation length a proper length.

The shared formal definitions are:

$$\ell_{E} \;=\; \hbar c/E$$

$$L \;=\; \ell_{E}\,\theta^{D}$$

$$x^{D} \;=\; Le^{a}$$

$$y^{D} \;=\; Le^{-a}$$

$$z^D \;=\; L$$

$$\Sigma \;=\; 2L(\cosh\; r \;-\; 1)$$

$$\theta^D \;\to\; \pi \;\to\; -@, \text{ with } r \;\to\; 0 \text{ and } \Sigma \;\to\; 0$$

15. D → A Transcription

At D closure:

$$r \;=\; 0$$

$$\Sigma \;=\; 0$$

$$L \;=\; L_{\text{cl}}$$

Define the completed D phase rate through the energy-frequency relation:

$$\omega^D \;=\; E^D/\hbar$$

PMM then adopts the $D \to A$ **phase-frequency continuity postulate** across $-@$:

$$\omega^A \;=\; \omega^D$$

therefore:

$$\omega^A \;=\; E^D/\hbar$$

and:

$$E^A \;=\; \hbar\omega^A$$

so:

$$E^A \;=\; E^D$$

Energy continuity is therefore **derived from** the $D \to A$ phase-frequency continuity postulate together with $E = \hbar\omega$; it is not an independent postulate.

The $D \to A$ phase-frequency continuity fixes the intrinsic unit-span transcription scale:

$$T^A{*} \;=\; 1/\omega^A$$

$$=\; \hbar/E^D$$

The physical A record-span time is:

$$T^A \;=\; T^A * \Delta\theta^A$$

$$= \Delta\theta^A/\omega^A$$

The event-level minimal A-record content is:

$$R^A_\alpha \;=\; \{\omega^A,\; a_{\rm cl},\; I^A\}$$

For a realized pair event α with endpoints on lineages i and j, C_3 produces two persistent record instances:

$$R^A_{i|\alpha} \text{ and } R^A_{j|\alpha}$$

with:

$$\text{Event}(R^A_{i|\alpha}) \;=\; \text{Event}(R^A_{j|\alpha}) \;=\; \alpha.$$

This is one C_3 transcription at one event closure with two endpoint writes, not two C_3 events or two FTUs.

Each endpoint instance contains the same event-level minimal core R^A_α unless a future PMM-native bridge derives endpoint-dependent readouts. Both endpoint instances reference the event-level energy $E^A(\alpha)$.

where:

$$a_{\rm cl} \;=\; \tfrac{1}{2}\ln(x^D_{\rm cl}/y^D_{\rm cl})$$

$$I^A \;=\; -\ln\; P_e@$$

The universal closure angle $\theta_{\rm cl} = \pi$ need not be stored event-by-event. From ω^A, PMM reconstructs $E^A = \hbar\omega^A$ and therefore $L_{\rm cl} = (\hbar c/E^A)\theta_{\rm cl}$. Together with the surviving planar-shape coordinate $a_{\rm cl}$, the completed x-y readout is reconstructible through $x^D_{\rm cl} = L_{\rm cl}\, e^{a_{\rm cl}}$ and $y^D_{\rm cl} = L_{\rm cl}\, e^{-a_{\rm cl}}$. Native z depth is not written into A.

C_3 is therefore :

what persists and is transcribed

Completed-event record. For realized pair event $\alpha = \{\text{i,j}\}$, the minimal numerical record is:

$$R^A_\alpha \;=\; \{\omega^A,\; a_{\rm cl},\; I^A\}.$$

C_3 supplies endpoint instances $R^A_{i|\alpha}$ and $R^A_{j|\alpha}$. These refer to one completed event and one event-level energy, not two completed events or independent source-energy copies. The multi-FTU update law (§31) appends the appropriate endpoint instance along its lineage.

At terminal vantage lock $\theta^D = \pi$, $V^D_1|\text{cl} = V^D_2|\text{cl} = (x^D_{\rm cl}, y^D_{\rm cl})$. This is a D-closure result, not an assertion that the full D trajectory is stored. Active xyz realization belongs to D; active depth ends at transcription.

From R^A_α, the existing equations reconstruct E^A, $L_{\rm cl}$, completed x-y scale and shape, and $\Delta\theta^A = I^A$. The D closure span is π; the candidate B span follows conditionally from $P_e@ = \exp(-I^A)$. These reconstructions do not establish retention or reconstruction of the original $\psi^B(x, y, z)$, arbitrary relative phase, or the full D trajectory. Any additional retained data require an explicit record specification and transcription law.

Interpretation (no new equation): completed records, which accumulate in Φ and condition gravity (§31), are completed-FTU imprints. The record and the gravity sector therefore look back on the same final FTU, and neither carries native depth.

16. A Phase — Persistent Record

After begins when active spatial realization closes. The event ceases its During activity and persists through its record.

Each lineage-endpoint A record stores an instance of completed-event information. Active xyz belongs to D; native depth z^D ends at D $\rightarrow$ A. Completed x-y scale and shape remain available through the existing R^A reconstruction. For event α, endpoint instances $R^A_{i|\alpha}$ and $R^A_{j|\alpha}$ refer to the same event-level record (§15).

Interpretation:

$$B \ \rightarrow \ \text{Wave}$$

$$D \ \rightarrow \ \text{Particle}$$

$$A \ \rightarrow \ \text{Cube}$$

Technical interpretation:

$$\text{Wave} \ = \ \text{possibility representation}$$

$$\text{Particle} \ = \ \text{realized spatial representation}$$

$$\text{Cube} \ = \ \text{completed-event representation}$$

The A cube is an event-record representation. B, D, and A also provide the three phase-temporal coordinates of F^T_{tu}. The current formalism does not impose an orthogonal Euclidean temporal-cube metric, and the retained x-y readout remains record rather than active spatial geometry after native depth z^D has ended.

Entropy belongs to the persistent A record. For the actualized pair e@, define:

$$I^A \ = \ -\ln \ P_e@$$

$$\Omega_A \ = \ e^{I^A} \ = \ 1/P_e@$$

$$\mathcal{S}^A \ = \ k_B I^A \ = \ k_B \ \ln \ \Omega_A$$

Ω_A is the effective A-side multiplicity of the realized event. PMM therefore has the dimensionless A information span:

$$\Delta\theta^A \ = \ \mathcal{S}^A/k_B \ = \ \ln \ \Omega_A \ = \ -\ln \ P_e@$$

The associated A phase time and action readouts are:

$$T^A \ = \ [\hbar/E^A]\Delta\theta^A \ = \ \Delta\theta^A/\omega^A$$

$$S^A \ = \ E^A T^A \ = \ \hbar\Delta\theta^A$$

The intrinsic unit-span transcription scale remains $T^A* = 1/\omega^A$; the physical A record span need not equal one unit cycle.

Status: mathematical event-surprisal readout; its physical A phase/action assignment is candidate PA-A, tied to $P_e@$ and carried through the stipulated transcription.

17. Geometry and Space

PMM distinguishes geometry from space.

Geometry spans B $\rightarrow$ D $\rightarrow$ A.

But:

ordinary xyz space is active during D.

Therefore:

B = pre-spatial geometry

D = spatial geometry

A = post-spatial geometry

The FTU is geometric throughout the event.

Only D carries active xyz spatial realization. B carries potential spatial content without active spatial coordinates. At A, the completed x-y spatial readout is retained as record rather than active space.

This is the internal geometry of one FTU. Shared physical space is assembled from relations among FTUs.

18. Single-FTU Vantage: Internal Geometry

The canonical single-FTU spine is:

B

$$\downarrow \Psi^B$$

$$C_1[\Phi]$$

$$\downarrow \text{ admissible pair-contact subspaces } \mathcal{H}_e^B[\Phi]$$

$$\downarrow Q_e \;=\; \text{Proj}(\mathcal{H}_e^B[\Phi])$$

$\downarrow P_e@$ = binary unordered-pair contact probability

$$@ \;:\; \text{actualize } e@ \;=\; \{i,j\};\ \text{assign event identifier } \alpha$$

$$\downarrow \text{ selected state } \Psi_e^B$$

$\downarrow$ candidate reciprocal channel weights $p_1|_e@ + p_2|_e@ = 1$

$$\downarrow P1 : \lim_{\theta^D \to 0^+} \rho_1/\rho_2 \;=\; p_1|_e@/p_2|_e@$$

$$\downarrow r_0 \;=\; \tfrac{1}{2}\ln(p_1|_e@/p_2|_e@)$$

$$\downarrow$$

D

Primitive D vantages:

$$c^1 \mapsto V_1 = (x_1^D, y_1^D)$$

$$c^2 \mapsto V_2 = (x_2^D, y_2^D)$$

$$\rho_1 = \sqrt{x_1^D y_1^D}$$

$$\rho_2 = \sqrt{x_2^D y_2^D}$$

Vantage geometry:

$$L = \sqrt{\rho_1 \rho_2}$$

$$r = \tfrac{1}{2} \ln(\rho_1/\rho_2)$$

$$L = \ell_E \theta^D$$

$$r'' + \tfrac{1}{4} r = 0$$

Vantage shape disagreement and common spatial readout:

$$b = \tfrac{1}{4} \ln[(x_1^D/y_1^D)/(x_2^D/y_2^D)]$$

$$x^D = \sqrt{x_1^D x_2^D}$$

$$y^D = \sqrt{y_1^D y_2^D}$$

$$a = \tfrac{1}{2} \ln(x^D/y^D)$$

$$z^D = L = \sqrt{x^D y^D}$$

$$\Sigma = \rho_1 + \rho_2 - 2z^D$$

$$\Theta_{\text{win}} = \sin(\pi q)$$

$$As\ \theta^D \to \pi :$$

$$r \to 0$$

$$\Sigma \to 0$$

$$\rho_1 \ \rightarrow \ \rho_2$$

$$-@$$

$$\downarrow$$

$$C_3$$

$$\downarrow \ \omega^A \ = \ E^D/\hbar; \ E^A \ = \ E^D$$

A

$$\downarrow \ R^A_{i|\alpha}, \ R^A_{j|\alpha} \text{ carrying } R^A_\alpha \ = \ \{\omega^A, a_{\text{cl}}, I^A\}$$

two endpoint persistent records / one event identity

19. Single-FTU Formal Status

PMM 3.7.6 presents the single-FTU vantage as a formal dependency structure. The ledger distinguishes **primitives, definitions/readouts,** postulates/**ansätze, imported physical relations, and conditional derivations**. Physical interpretations are not established by assigning a mathematical readout.

19.1 Primitives

B, D, A — phase architecture

$$@, \ -@ \text{ — phase boundaries}$$

$$\theta^D \text{ — angular } D \text{ progression}$$

$$E^D \text{ — } D \text{ carrier energy}$$

c^1, c^2 — D realization grades

$$V_1 \ = \ (x^D_1, y^D_1), \ V_2 \ = \ (x^D_2, y^D_2) \text{ — two } D \text{ spatial vantages}$$

These form the starting objects of the single-FTU construction.

Imported physical relations used in this construction: E = $\hbar\omega$ for the stipulated carrier phase-frequency readout; $E^D = mc^2$ in the massive rest-energy sector; and $\ell_p^2 = \hbar G_N/c^3$. These are adopted standard-physics inputs, not derived PMM first principles.

19.2 Definitions and Readouts

$$\rho_1 \;=\; \sqrt{x_1^D y_1^D}$$

$$\rho_2 \;=\; \sqrt{x_2^D y_2^D}$$

$$\ell_{_E} \;=\; \hbar c/E$$

$$L \;=\; \sqrt{\rho_1 \rho_2}$$

$$r \;=\; \tfrac{1}{2}\ln(\rho_1/\rho_2)$$

$$x^D \;=\; \sqrt{x_1^D x_2^D}$$

$$y^D \;=\; \sqrt{y_1^D y_2^D}$$

$$a \;=\; \tfrac{1}{2}\ln(x^D/y^D)$$

$$b \;=\; \tfrac{1}{4}\ln[(x_1^D/y_1^D)/(x_2^D/y_2^D)]$$

$$z^D \;=\; L$$

$\rho_+^D \;=\; \frac{1}{2}(\rho_1 \;+\; \rho_2)$ — symmetric D-vantage readout

$\rho_-^D \;=\; \frac{1}{2}(\rho_1 \;-\; \rho_2)$ — antisymmetric D-vantage readout

$$\Sigma \;=\; \rho_1 \;+\; \rho_2 \;-\; 2z^D$$

$$q \;=\; \theta^D/\theta_{\mathrm{cl}}$$

$X^D \;=\; q \;=\; \theta^D/\pi$ — normalized D expansion readout

$C^D \;=\; \Theta_{\mathrm{out}} \;=\; \cos(\theta^D/2)$ — normalized D contraction readout

$\tilde{\Xi}^D = (\mathrm{r},\, -2r')^T$ — derived unnormalized two-component half-angle carrier; preserves the magnitude and sign of r_0

$$\omega^D \;=\; E^D/\hbar$$

$\mathrm{F}_{tu}^M,\ \mathrm{F}_{tu}^E,\ \mathrm{F}_{tu}^T$ — primary FTU readouts

$\mathrm{F}_{tu}^{xy} \;=\; (\psi(x,y,z),\ \mathrm{xyz},\ xy)$ — spatial readout

Active spatial dimensionality = (0, 3, 0) across B, D, A — active-space readout

$$\mathrm{F}^{H}_{tu} \;=\; (B,(B,D),(B,D,A)) \text{ — cumulative phase-history readout}$$

$\mathrm{F}^{\mathrm{Hxy}}_{tu} = (B,\psi(x,y,z)) \to ((\mathrm{B,D}),\mathrm{xyz}) \to ((\mathrm{B,D,A}),\mathrm{xy})$ — combined phase-history / spatial readout

$\mathrm{F}^{s}_{tu} = S^B S^D S^A = \hbar^3 \Delta\theta^B \theta^D \Delta\theta^A$ — candidate Action Cube, units $(\mathrm{J}\cdot\mathrm{s})^3$

$\mathrm{F}^{\mathrm{led}}_{tu} = S^B + S^D + S^A$ — additive Action Ledger, units $\mathrm{J}\cdot\mathrm{s}$

$\mathrm{F}^{sv}_{tu} = (S^B, S^D, S^A)$ — Phase-Action Vector, units $\mathrm{J}\cdot\mathrm{s}$ per coordinate

$$dS^D \;=\; \hbar d\theta^D \text{ — } D \text{ angular-action normalization}$$

A mathematical information/readout definitions and candidate physical assignments:

$$I^A \;=\; -\ln\, P_e@$$

$$\Omega_A \;=\; 1/P_e@$$

$$\Delta\theta^A \;=\; \mathcal{S}^A/k_B \;=\; \ln\, \Omega_A \;=\; I^A$$

$$S^A \;=\; \hbar\Delta\theta^A$$

These introduce notation and read out specified states. Because both $\Delta\theta^B = \arccos\sqrt{P_e@}$ and $\Delta\theta^A = -\ln\, P_e@$ depend on the same selected-pair probability, the B and A spans are explicitly coupled through $P_e@$. F^{H}_{tu} records completed event history, while $\mathrm{F}^{\mathrm{Hxy}}_{tu}$ displays that history together with the existing spatial readout. Both are distinct from F^{T}_{tu}, which is the three-coordinate temporal state-space representation. The temporal-state-space statement does not require the three coordinates to be orthogonal or freely independent.

19.3 Explicit Single-FTU Postulates / Ansätze

P0 c-order correspondence:

$$B \;=\; (c^0)_{mp}$$

$$D \;=\; (c^1, c^2)c$$

$$A \;=\; (c^3)k_B$$

PA-A candidate A information-to-phase/action assignment:

$$\Delta\theta^A \;=\; I^A \;=\; -\ln\, P_e@$$

$$\mathcal{S}^A \;=\; k_B I^A;\; S^A \;=\; \hbar I^A$$

The event surprisal is a definition; assigning it a physical A phase/action span is candidate physical content, not a derivation of thermodynamic entropy.

GC-A candidate gravitational-crease closure condition:

$A_{\mathrm{cl}} = A_G = \ell_p^2$ at the proposed limiting boundary only

Native C_3 selection of this equality remains open.

PA-B candidate B projective action span:

$$\Delta\theta^B \;=\; \arccos\sqrt{P_e}@$$

$$S^B \;=\; \hbar\Delta\theta^B$$

C_1-B candidate reciprocal pair-contact probability :

$$P_e \;=\; |a_e|^2 \;/\; \sum_k |a_k|^2$$

C_1-C candidate pair-contact construction for the Pairwise Realization Postulate of §20.1:

$$C_1[\Phi] \;\to\; \mathcal{H}_e^B[\Phi]$$

$$Q_e \;=\; \mathrm{Proj}(\mathcal{H}_e^B[\Phi])$$

$$P_e@ \;=\; \langle \Psi_a^B d_m | Q_e | \Psi_a^B d_m \rangle$$

$$P_e@ \;+\; P(\neg e@) \;=\; 1$$

$$P_{ij}@ \;=\; P_{ji}@ \;=\; P_e@$$

$\sum P_e@ = 1$ only inside a complete mutually exclusive C_1 contact block

P1 $B \to D$ ratio transfer within the selected pair e@:

$$\lim_{\theta^D \to 0^+} \;\rho_1/\rho_2 \;=\; p_1|_e@/p_2|_e@$$

P2 D scale law:

$$L \;=\; \ell_E\, \theta^D$$

P3 terminal vantage lock:

$-@ \;\Leftrightarrow\; \theta^D \;=\; \pi \;\Rightarrow\; r \;=\; 0,\; b \;=\; 0,\; \Sigma \;=\; 0$

```
the two c-grade vantages agree in scale and planar shape at lock; during D they may disagree
```

P4 angular closure:

$$\theta_{\mathrm{cl}} \;=\; \pi$$

P5 minimal linear conservative closure class:

$$r'' \;+\; \kappa_{\mathrm{cl}}\, r \;=\; 0$$

with $r'(0) = 0$ as the entry-symmetry condition

and, for $r_0 \neq 0$, the first zero at π

$$\Rightarrow \ \kappa_{\text{cl}} \ = \ \tfrac{1}{4}$$

P6 lowest-mode symmetric space window:

zero endpoints, positive interior, midpoint symmetry, unit peak

plus no internal nodes / no extra oscillations

$$\Rightarrow \ \Theta_{\text{win}} \ = \ \sin(\pi q)$$

$$\text{with } \Theta_{\text{in}} \ = \ \sin(\pi q/2), \ \Theta_{\text{out}} \ = \ \cos(\pi q/2)$$

$$P7 \ D \to A \text{ phase-frequency continuity :}$$

$$\omega^A \ = \ \omega^D$$

These are the irreducible assumptions and conditional-selection principles that govern the single-FTU vantage. The D angular-action relation is classified as a definition/readout in §19.2 rather than as an additional postulate. PA-A and GC-A are candidate physical assignments/conditions; C_1-B and C_1-C are candidate entry rules rather than derived laws. P5 and P6 are selected within their adopted minimal admissible classes.

19.4 Derived Results

From the definitions and postulates above:

$$r_0 \ = \ \tfrac{1}{2} \ln(p_1|_e @ / p_2|_e @)$$

$$\rho_1 \ = \ Le^r$$

$$\rho_2 \ = \ Le^{-r}$$

$$x^D \ = \ Le^a$$

$$y^D \ = \ Le^{-a}$$

$$x^D y^D \ = \ L^2$$

$$z^D \ = \ L \ = \ \sqrt{x^D y^D}$$

$$\rho_+^D \ = \ L \ \cosh \ r$$

$$\rho_-^D \ = \ L \ \sinh \ r$$

$$(\rho_+^D)^2 \ - \ (\rho_-^D)^2 \ = \ L^2 \ = \ (z^D)^2$$

under $\rho_1 \rightarrow \mu\rho_1$ and $\rho_2 \rightarrow \rho_2/\mu : r \rightarrow r + \ln \mu$ while L and z^D remain invariant

$$\Sigma = 2L(\cosh r - 1)$$

$$= 4L \sinh^2(r/2)$$

$$\Sigma \geq 0$$

For L > 0: $\Sigma = 0 \Leftrightarrow$ r $= 0 \Leftrightarrow \rho_1 = \rho_2 = z^D$

$$\kappa_{\text{cl}} = \tfrac{1}{4}$$

$$r(\theta^D) = r_0 \cos(\theta^D/2)$$

$$\tilde{\Xi}^D(\theta^D) = r_0(\cos(\theta^D/2), \sin(\theta^D/2))^T$$

$$\tilde{\Xi}^D(0) = (r_0, 0)^T; \ \tilde{\Xi}^D(\pi) = (0, r_0)^T$$

$\tilde{\Xi}^D(2\pi) = -\tilde{\Xi}^D(0)$; $\tilde{\Xi}^D(4\pi) = \tilde{\Xi}^D(0)$ as the mathematical continuation of the half-angle harmonic law

$$X^D = q = \theta^D/\pi : 0 \rightarrow 1$$

$$C^D = \Theta_{\text{out}} = \cos(\theta^D/2) : 1 \rightarrow 0$$

$$L_{\text{cl}} = \pi\hbar c/E^D$$

$$\Theta_{\text{win}} = \sin(\pi q)$$

$$E^A = E^D$$

$$I^A = -\ln P_e@$$

$$\Omega_A = 1/P_e@$$

$$\Delta\theta^A = I^A = \ln \Omega_A$$

$$\mathcal{S}^A = k_B I^A$$

$$S^A = \hbar I^A$$

$$T^{A}{}_{*} \;=\; \hbar/E^{D}$$

$$S^{D}(\theta^{D}) \;=\; \hbar\theta^{D}$$

$$S^{D},\mathrm{cl} \;=\; \pi\hbar$$

$$\mathrm{F}^{\mathrm{led}}_{tu},\mathrm{cl} \;=\; \hbar(\Delta\theta^{B} \;+\; \pi \;+\; \Delta\theta^{A})$$

The distinct candidate coordinate product at the same closure is:

$\mathrm{F}^{s}_{tu},\mathrm{cl} = \pi\hbar^{3}\Delta\theta^{B}\Delta\theta^{A}$, units $(\mathrm{J}\cdot\mathrm{s})^{3}$.

This is an algebraic evaluation of the candidate definition, not an independently derived physical volume or a replacement for the action ledger.

The coefficient $\frac{1}{4}$ follows from P4 plus the harmonic form in P5. Depth follows from the two-vantage construction. Energy continuity follows from P7 plus the energy-frequency relation. The Expansion–Contraction Readout follows from P2 together with the derived harmonic closure solution. The unnormalized two-component half-angle equation for $\tilde{\Xi}^{D}$ is an exact first-order rewriting of the same harmonic law. Its 4π period is a property of the half-frequency oscillator; a physical spinor interpretation requires additional rotational generators and a physical orientation bridge.

The standard identity $\ell_{p}^{2} = \hbar G_{N}/c^{3}$ appears in the c-order profile and in the local c^{3} discussion. The candidate gravitational-crease condition $A_{\mathrm{cl}} = A_{G}$, when applied to the current π closure geometry, yields $E^{D}_{\mathrm{cl}} = \pi m_{p}c^{2}$; this remains a candidate limiting condition pending a native C_{3} selection law.

19.5 Massive / Null Consistency

The single-FTU geometry includes both carrier sectors:

massive:

$$E^{D} \;=\; mc^{2}$$

$$\ell_{E} \;=\; \hbar/(mc)$$

proper time accumulates

null:

$$E\gamma \;=\; \hbar\omega$$

$$\ell_{E} \;=\; c/\omega$$

$$d\theta^{D} \;=\; \omega dt$$

$$dL\gamma/dt \;=\; c$$

$$d\tau^{D} \;=\; 0$$

$$dS^{D} \;=\; \hbar d\theta^{D}$$

The null sector has its own native definition. The massive and null sectors share the same L, r, a, z^{D}, Σ geometry, angular-action readout, and closure condition.

19.6 Minimality / Open Single-FTU Work

The current model structurally specifies the single-FTU architecture. Remaining work focuses on reducing its postulates:

OPEN derive or reduce P0, the c-order correspondence, from deeper PMM structure if possible

OPEN derive or reduce the C_1 pair-contact probability rule from deeper PMM structure

OPEN test the candidate reciprocal pair-contact probability and pair-basis invariance

OPEN test or derive the candidate B projective span $\Delta\theta^B = \arccos\sqrt{P_e}@$

OPEN derive or reduce P1 if a deeper $B \to D$ principle is found

OPEN derive or reduce P2 if angular time uniquely fixes the D scale law

OPEN derive P3 (terminal vantage lock) from the closure structure, and determine whether b follows a closure law such as $b'' + \frac{1}{4}b = 0$

OPEN determine whether the vantage-shape disagreement b couples to the shape a; the only quadratic source with matching exchange charges is $r \cdot b$ (candidate, not adopted)

OPEN derive P4 from a deeper closure principle

OPEN test whether the minimal linear conservative class in P5 is itself forced

OPEN test whether the lowest-mode condition in P6 is itself forced

OPEN derive or reduce P7 if transcription continuity follows from C_3 itself

OPEN determine whether Θ_{win} is purely an active-space access readout or whether it also enters the phase-action ledger; the current $\mathrm{F}_{tu}^{\text{led}}$ ledger is unwindowed; separately determine whether Θ_{win} affects the candidate Action Cube readout

OPEN determine whether the native D geometry supplies additional noncommuting generators beyond the real half-angle carrier

OPEN determine whether θ^D progression can be physically identified with rotational orientation in the multi-FTU sector

OPEN determine how the magnitude and sign carried by $\tilde{\Xi}^D$ could persist through A $\to \Phi \to$ later FTUs, if such persistence is physical

OPEN derive or reject the physical identification of V_1 and V_2 as reciprocal null/light-channel vantages of one D interaction

19.7 Canonical Single-FTU Spine

B potential Ψ^B

$\downarrow$

C_1-C: $C_1[\Phi]$ defines admissible pair-contact subspaces $\mathcal{H}_e^B[\Phi]$

$\downarrow$

$$Q_e \;=\; \text{Proj}(\mathcal{H}_e^B[\Phi]) \text{ and binary } P_e@$$

$\downarrow$

@ actualizes e@ = {i,j}; event identifier assigned after contact

$\downarrow$

selected B state Ψ_e^B

$$\downarrow$$

candidate reciprocal D-vantage weights $p_1|_e@,\ p_2|_e@$

$$\downarrow$$

$$P1:\ p_1|_e@/p_2|_e@\ \rightarrow\ \lim_{\theta^D\rightarrow 0^+}\ \rho_1/\rho_2\ =\ e^{2r_0}$$

$$\downarrow$$

$$c^1\ \mapsto\ V_1\ =\ (x_1^D, y_1^D)$$

$$c^2\ \mapsto\ V_2\ =\ (x_2^D, y_2^D)$$

$$D \text{ action definition}:\ dS^D\ =\ \hbar d\theta^D$$

$$\downarrow$$

$$\rho_1,\ \rho_2\ \rightarrow\ L,\ r\ \rightarrow\ \rho_+^D,\ \rho_-^D \text{ with } (\rho_+^D)^2\ -\ (\rho_-^D)^2\ =\ L^2$$

$$P2:\ L\ =\ \ell_E\theta^D$$

derived D pair: $X^D = \mathrm{q}$ rises $0 \rightarrow 1$ while $C^D = \Theta_{\text{out}}$ falls $1 \rightarrow 0$

P3: terminal vantage lock (r = 0, b = 0 at $-@$); common x^D, y^D and $z^D = \mathrm{L}$ hold identically from the four-mode decomposition

P4/P5: π closure + minimal linear conservative class $\Rightarrow \kappa_{\text{cl}} = \frac{1}{4}$

$$P6:\ \text{lowest-mode window}\ \Rightarrow\ \Theta_{\text{win}}\ =\ \sin(\pi q)$$

$$\downarrow$$

$$r\ \rightarrow\ 0,\ \Sigma\ \rightarrow\ 0,\ \rho_1\ \rightarrow\ \rho_2$$

$$\downarrow$$

$$-@$$

$$\downarrow$$

$$P7: \ \omega^A \ = \ \omega^D$$

$$\downarrow$$

A persistent record

This is the canonical single-FTU dependency chain for PMM 3.7.6.

19.8 Recovery Methodology / Firewall

PMM keeps native construction separate from recovery. A recovery begins with PMM primitives and derived relations, applies a stated limit or bridge, and then compares the result with established physics.

A successful recovery match establishes compatibility within its stated limit. Adoption as a first principle requires a separate PMM-native derivation or explicit postulate.

19.9 Candidate Test / Falsification Discipline

Every candidate states its dependencies, test conditions, and failure condition. Failure rejects or revises the candidate; passage establishes consistency within the tested domain. First-principle adoption requires a separate derivation or explicit postulate.

Promotion requires a PMM-native derivation, or a new independently testable consequence that survives its stated falsification test.

PART III — MULTI-FTU VANTAGE

20. Multi-FTU Principle

A complete FTU becomes part of a continuing physical history. Each shared pairwise event contains two internal During vantages and writes endpoint records under one event identity. Across later contacts, participant lineages connect these records into an evolving network. Concurrent contacts belong to distinct events even when no causal order separates them; participant count and event count remain separate.

The multi-FTU vantage describes the relations among these events, their retained history, and their clock responses. It extends the same BDA process beyond one completed FTU.

The two are separate sectors of one theory:

single FTU

→ local event geometry

multi FTU

→ shared relational geometry

They couple through the event-record cycle:

single FTU

→ two endpoint A records / one event identity

$$\rightarrow \ \text{multi-FTU field}$$

→ future C_1 / C_2 conditions

$$\rightarrow \ \text{later single FTUs}$$

Thus PMM is one recursive theory expressed through two vantages.

20.1 Pairwise Realization and Lineage Continuity

Pairwise Realization Principle

Elementary realized contact is pairwise. Each elementary physically actualized unordered pair contact:

$$e@ \;=\; \{i, j\}$$

constitutes one FTU.

Before realization, $C_1[\Phi]$ specifies the admissible pair-contact propositions Q_e. Exclusive alternatives compete inside complete orthogonal blocks; compatible commuting contacts admit concurrent actualization. Event identifiers α, β, ... are assigned only after actualization and distinguish completed or active FTUs from participant identity.

For realized FTU α:

$$e@(\alpha) \;=\; \{i, j\}.$$

The FTU has two endpoint occurrences, one associated with each participating carrier/history branch. Let $h(i|\alpha)$ denote the endpoint occurrence associated with lineage i in event α.

Lineage Continuity Principle

> Two endpoint occurrences belong to the same participant lineage when they are connected by the native continuation relation $\sim^L$ across successive compatible FTU events.

A participant lineage $\mathcal{P}_i$ is the continuity class generated by $\sim^L$ across its incident FTUs. Participant identity is therefore not identified with c^1, c^2, V_1, or V_2.

The internal D slots remain ordered:

$$c^1 \;\mapsto\; V_1$$

$$c^2 \;\mapsto\; V_2$$

with candidate conditional seed weights:

$$p_1|_e@ \;+\; p_2|_e@ \;=\; 1$$

and:

$$r_0 \;=\; \tfrac{1}{2}\ln(p_1|_e@/p_2|_e@).$$

The indices 1 and 2 order reciprocal internal D-vantage slots; i and j identify participant lineages. Pair selection assigns no intrinsic c^1/c^2 identity, positive identity, or negative identity to either participant. Exchanging V_1 and V_2 reverses r while preserving L, z^D, and Σ.

An external participant k describes a completed event; physical contact determines event count. If k physically measures or otherwise contacts i, j, or a physical carrier of the event record, that physical interaction is a new pairwise @ and therefore a new FTU.

If two compatible external contacts actualize without causal/contact order between them, PMM counts two distinct FTUs in the same relational slice. Concurrency does not merge their V_1/V_2 geometry, action ledgers, closures, or A records.

Multi-FTU geometry is assembled from compatible realized pair relations across relational history. Three-participant loops and four-participant relational volume therefore require no primitive many-body FTU.

Status: Pairwise Realization and Lineage Continuity are EXTENSION POSTULATES of the multi-FTU sector. The deeper dynamical derivation of $\sim^L$, including creation, termination, merger, or branching of carrier histories, remains OPEN. The native law $C_1[\Phi] \to \mathcal{H}_e^B[\Phi]$ also remains OPEN / CANDIDATE.

20.1.1 FTU∞ Relational Contact Construction

Pair contact brings two participating lineages into one realized event. Each participant contributes a traversal input to the shared contact. The single-FTU vantage describes the internal evolution of that event; the multi-FTU vantage connects its participating lineages and completed records across subsequent events.

FTU∞ denotes the exchange decomposition of two contact inputs. For traversal vectors u_i and u_j, define the unnormalized exchange-even and exchange-odd sectors:

$$\tilde{S}_{ij} \;=\; \tfrac{1}{2}(u_i \otimes u_j + u_j \otimes u_i)$$

$$\tilde{A}_{ij} \;=\; \tfrac{1}{2}(u_i \otimes u_j - u_j \otimes u_i)$$

Exchanging the participants gives:

$$\tilde{S}_{ji} \;=\; \tilde{S}_{ij}$$

$$\tilde{A}_{ji} \;=\; -\tilde{A}_{ij}$$

The exchange-even sector carries shared relational content; the exchange-odd sector carries orientation information. When both traversal inputs are expressed in a common length-valued space, $\tilde{S}$ and $\tilde{A}$ have dimension L^2. A local realization map converts their content into the corresponding length-valued D-vantage readouts:

$$(V_1, V_2) \;=\; \boldsymbol{\mathcal{R}}_e(\tilde{S}_{ij}, \tilde{A}_{ij};\; \chi_e)$$

Here χ_e collects the event's already-defined contact, scale, and orientation data. The map $\boldsymbol{\mathcal{R}}_e$ is an explicit mathematical interface between the historical FTU∞ decomposition and the current two-vantage construction; no identity with the current D geometry is presumed. The resulting V_1, V_2 enter the existing relations of §§6–7, including $\mathrm{L} = \sqrt{\rho_1 \rho_2}$ and $z^D = \mathrm{L}$, without altering those definitions.

Mathematical status: The symmetric/antisymmetric exchange identities are DERIVED from the displayed tensor definitions. The specific realization map $\boldsymbol{\mathcal{R}}_e$ is CANDIDATE; its specification and comparison with the current D-vantage data are OPEN. This contact construction does not introduce an additional FTU, contact, closure, or record. Each physically actualized unordered pair contact continues to count as one event under §20.1.

Verification condition: An admissible $\boldsymbol{\mathcal{R}}_e$ preserves participant-exchange invariants, reverses orientation where required, reproduces the positive D-vantage scales, and respects the existing closure identities. A failure of these conditions rejects that map while leaving the established §20.1 event accounting and §§6–7 geometry unchanged.

20.2 M_{op} — Motion Operations Principle

$\mathbf{M_{op}}$ — Motion Operations Principle

Space is emergent relational motion. Space and motion are two aspects of the same realized D relation: a realized relation supplies spatial separation, and evolution of that relation supplies motion. Without a realized relation between lineages, there is no realized spatial separation and no relational motion between them.

Across the FTU architecture:

B = potential relation / potential motion / pre-spatial possibility

D = realized relation / relational motion / active space

A = completed relation / persistent record of the realized event

The realization crease c is the invariant still reference of D. PMM therefore reads null propagation through the existing relation $dL\gamma/dt = \mathrm{c}$ with $d\tau^D = 0$: the photon is still in the zero-proper-time sense while D realization propagates. M_{op} introduces no photon rest frame.

For an inter-participant relation δ_{ij}, relational motion is the ordered evolution of that edge across relational history. Using the existing comparison time t, define the relational-motion readouts:

$$v_{ij}^R \equiv d\delta_{ij}/dt$$

$$a_{ij}^R \equiv d^2\delta_{ij}/dt^2.$$

Their roles are distinct:

outwardness / inwardness = orientation of relational evolution

acceleration / deceleration = change in the rate of relational evolution

directionality = orientation reconstructed from the embedded δ network

Thus:

$$v_{ij}^R > 0 \rightarrow \text{outward relational evolution}$$

$$v_{ij}^R < 0 \rightarrow \text{inward relational evolution}$$

and, for nonzero v_{ij}^R:

$$v_{ij}^R\, a_{ij}^R > 0 \rightarrow \text{relational motion speeds } up$$

$$v_{ij}^R\, a_{ij}^R < 0 \rightarrow \text{relational motion slows down.}$$

Acceleration is therefore distinct from outwardness: an outward relation can slow while an inward relation can speed up.

At the discrete level, the ordered sequence $\delta_{ij}^n \rightarrow \delta_{ij}^{n+1} \rightarrow \delta_{ij}^{n+2}$ carries the same content: first relational change is motion and change of that relational rate is acceleration/deceleration. The continuum derivatives above are the comparison-time readouts of that discrete evolution.

The gravity sector supplies $g = c^2\nabla\varphi$. Under M_{op}, a common acceleration of neighboring lineages changes their common motion without producing local relational deformation, while differential acceleration changes their δ relations and therefore deforms the assembled relational geometry. After embedding, the edge-projected relative gravitational acceleration is read as:

$$a_{ij}^R = (g_j - g_i) \cdot \hat{n}_{ij}$$

where $\hat{n}_{ij}$ is reconstructed from the embedded relational geometry rather than introduced as a primitive global direction. Curvature remains the assembled δ-network readout of §23. M_{op} supplies the dynamical interpretation: nonuniform relational motion changes the relations from which spatial geometry is assembled.

M_{op} fixes the ontology and kinematic hierarchy. The full nonlinear evolution law coupling κ, φ, δ, L, a, and cross-FTU orientation remains a derivation target.

21. Primitive Relational Geometry

The core relational quantity is inter-participant distance:

$$\delta \;=\; \text{inter-participant distance}$$

For a realized pair of participants e@ = {i,j}, the relational edge is:

$$\delta_{ij} \;=\; \delta_{ji}$$

$$\delta_{ii} \;=\; 0$$

$$\delta_{ij} \;>\; 0 \text{ for } i \;\neq\; j$$

The relational distance matrix is:

$$M\delta \;=\; \{\delta_{ij}\}$$

The δ indices denote participant lineages, not separately counted FTU events. No global direction vector is required as a primitive. Direction can be reconstructed after the distance network is embedded.

The distinction is:

z^D = internal single-FTU D depth (between the two x-y vantages of one FTU)

δ_{ij} = inter-participant distance (between two participants, not between vantages)

δ_{ij} is an **independent primitive of the multi-FTU sector**. The direct bridge between single-FTU depth z^D and inter-participant distance δ is an open derivation target (see §7.4).

The two vantages couple through completed FTU outputs and the shared history/source sector; the direct $z^D \leftrightarrow \delta$ bridge is open.

22. Shared Space from FTU Relations

One FTU gives an internal event view.

Two participants give a separation through a shared FTU.

Three compatible participants give the first closed relational loop through pairwise FTUs.

Four suitably independent participants give relational three-dimensional volume.

$$1 \text{ FTU} \;\rightarrow\; \text{internal event view}$$

$$2 \text{ participants} \;\rightarrow\; \text{relational separation}$$

$$3 \text{ participants} \;\rightarrow\; \text{relational area / loop}$$

$$4 \text{ participants} \;\rightarrow\; \text{relational } 3D \text{ volume}$$

Pairwise δ_{ij} relations assemble shared physical space. These relations may be accumulated through successive or concurrent pairwise actualizations; a three- or four-participant geometric configuration does not require a primitive many-body contact.

The single-FTU architecture is ordinal and angular, while many overlapping FTU relations build shared physical geometry. PMM treats an effectively smooth macroscopic spacetime as a candidate recovery: a many-FTU coarse-graining bridge is required to connect the relational network with a continuum description at observable scales. The continuum limit and bridge are OPEN.

For four participants, the six consistent pair distances define a tetrahedral relational cell. Such cells assemble larger spatial geometry.

23. Discrete Curvature

When relational tetrahedra assemble around an edge, curvature can be represented by the deficit angle:

$$\Delta_e \;=\; 2\pi \;-\; \sum_t \theta_{te}$$

where θ_{te} is the dihedral angle contributed by tetrahedron t around edge e.

Then:

$$\Delta_e \;=\; 0 \;\rightarrow\; \text{locally flat assembly}$$

$$\Delta_e \;\neq\; 0 \;\rightarrow\; \text{relational curvature}$$

Curvature emerges from the assembled δ geometry.

23.1 Recovery Bridge

The discrete relational network supplies the native multi-FTU geometry. Standard differential geometry then provides a recovery bridge from that network to continuum curvature:

δ-network $\rightarrow$ local frame $\rightarrow$ connection $\rightarrow$ curvature $\rightarrow$ R

This bridge belongs to recovery. Standard frame, connection, and curvature tensors act downstream of the PMM relational structure rather than as PMM first principles.

24. Relational Event Lattice

A multi-FTU state at relational slice n is:

$$\boldsymbol{\mathcal{M}}^n \;=\; \{\delta_{ij}^n,\; \Phi_i^n,\; \kappa_i^n\}$$

or equivalently:

$$\boldsymbol{\mathcal{M}}^n \;=\; \{\delta_{ij}^n,\; \Phi_i^n,\; \varphi_i^n\}$$

with:

$$\varphi_i^n \;=\; -\ln\, \kappa_i^n.$$

Here $\mathcal{P}_i^n$ denotes lineage i at slice n, derived from endpoint continuity under $\sim^L$ rather than identified with a complete FTU. Each realized FTU has an event identifier α distinct from its participant pair e = {i,j}; repeated contacts of the same pair are distinct events.

The network therefore contains three typed structures:

relational edge:

$$\mathcal{P}_i^n \;\leftrightarrow\; \mathcal{P}_j^n$$

$$\text{length} \;=\; \delta_{ij}^{n}$$

lineage continuation:

$$\mathcal{P}_i^n \;\rightarrow\; \mathcal{P}_i^{n+1}$$

$$\text{history} \;=\; \Phi_i^n \;\rightarrow\; \Phi_i^{n+1}$$

lineage clock state:

$$\kappa_i^n \text{ or } \varphi_i^n.$$

Repeated relational slices produce an effective event geometry:

$$\boldsymbol{\mathcal{M}}^0 \;\rightarrow\; \boldsymbol{\mathcal{M}}^1 \;\rightarrow\; \boldsymbol{\mathcal{M}}^2 \;\rightarrow\; ...$$

Under M_{op}, the ordered change of δ across these slices is relational motion; change of that relational rate is acceleration/deceleration. A constant δ edge is a realized spatial relation with zero relative edge motion, not an absence of space.

Let $\mathcal{A}^n$ denote the set of distinct realized FTU event identifiers represented in relational slice n:

$$\mathcal{A}^n \;=\; \{\alpha,\; \beta,\; ...\}.$$

For each $\alpha \in \mathcal{A}^n$, one elementary pair has actualized:

$$e@(\alpha) \;=\; \{i, j\}.$$

If two events are causally ordered, the lattice records that order through the relevant lineages. If no causal/contact order is imposed between compatible events, they may coexist in the same relational slice as concurrent distinct FTUs. A relational slice may therefore contain several newly realized pairwise events together with relations accumulated from earlier slices while the primitive FTU remains pairwise.

Event-local quantities carry event identity, for example:

$$\theta^D(\alpha),\; E^D(\alpha),\; L(\alpha),\; r(\alpha),\; \Sigma(\alpha),\; R_\alpha^A.$$

The persistent endpoint instances $R_{i|\alpha}^A$ and $R_{j|\alpha}^A$ carry both lineage-endpoint identity and the same event identity α.

Lineage quantities carry participant identity:

$$\Phi_i,\; \kappa_i,\; \varphi_i.$$

Relational-edge quantities carry pair identity:

$$\delta_{ij}.$$

This typed separation is the PMM route from local FTU geometry to emergent relational spacetime.

25. A-Side Record and Recursive History

Each completed pair FTU writes one event into physical history. Its two lineage endpoints retain corresponding A-record instances, both referring to the same event identity α. Lineage identity is supplied by $\sim^L$; Φ stores the endpoint record referenced along that lineage but does not itself define carrier identity. When lineage i references one completed event α in a slice, the history update is:

$$\Phi_i^{n+1} \;=\; \mathrm{Append}(\Phi_i^n,\; R^A_{i|\alpha(i,n)})$$

For the general case, let $\mathcal{A}_i^n$ be the set of distinct completed FTU event identifiers referenced by participant i in relational slice n. Then:

$$\Phi_i^{n+1} \;=\; \mathrm{Append}(\Phi_i^n,\; \{R^A_{i|\alpha} \;:\; \alpha \;\in\; \mathcal{A}_i^n\})$$

The singleton case is recovered when $|\mathcal{A}_i^n| = 1$. For one shared event $\alpha = \{\mathrm{i,j}\}$, lineage i appends $R^A_{i|\alpha}$ and lineage j appends $R^A_{j|\alpha}$. These are distinct physical record instances of the same event, so Append must preserve endpoint identity while source extraction deduplicates by α. Record multiplicity therefore does not create independently duplicated source energy. The event-level minimal physical record content R^A_α is unchanged. Any endpoint-dependent physical readout beyond record identity requires a separate PMM-native derivation. A gravitational extraction rule must distinguish event identity, endpoint-record identity, concurrent records, repeated histories, and current source content; that rule remains open.

For the gravity sector, Φ_s denotes a scalar readout of the accumulated record Φ. Ordinary addition applies to Φ_s only in the reduced regime of §30.1; the general Append rule is not assumed to be additive.

The core constraint set already couples Φ back into later events:

$$\Theta_{\mathrm{con}} \;=\; \{$$

$$C_1[@, \Phi],$$

$$C_2[\Phi, \delta, \Sigma],$$

$$C_3[\Phi, \Sigma]$$

}

So the larger PMM cycle is:

$$B \;\to\; D \;\to\; A \;\to\; \Phi \;\to\; \text{future } B/D$$

Interpretation:

$C_1 = \Phi$ conditions future contact admissibility

$C_2 = \Phi$ conditions realized D propagation

$$C_3 \;=\; \text{completed } D \text{ writes persistent } A \text{ content}$$

Φ is therefore the memory channel by which completed geometry conditions future realization.

25.1 Candidate Relational Half-Angle Transport

The single-FTU D harmonic law supplies the real half-angle carrier:

$$\tilde{\Xi}^D(\theta^D) \;=\; r_0(\cos(\theta^D/2),\ \sin(\theta^D/2))^T$$

while canonical single-FTU closure supplies:

$$\theta_{\mathrm{cl}} \;=\; \pi$$

At closure:

$$\tilde{\Xi}^D(\pi) \;=\; (0, r_0)^T$$

so each nondegenerate FTU ends a quarter turn from its entry orientation in the internal carrier plane.

A cross-FTU transport law requires an additional continuity hypothesis. If one relational carrier is preserved coherently across completed FTU boundaries, define the planar transporter:

$$U(\theta) \;=\; \exp[(\theta/2)\mathrm{J}]$$

with:

$$\mathrm{J} \;=\; [[0,-1],[1,0]]$$

and the conditional history:

$$H_n \;=\; [U(\pi)]^n$$

Then:

$$H_1 \;=\; U(\pi)$$

$$H_2 \;=\; U(2\pi) \;=\; -I$$

$$H_3 \;=\; U(3\pi)$$

$$H_4 \;=\; U(4\pi) \;=\; I$$

This establishes 4π periodicity of the transported internal carrier under the continuity hypothesis. Calling the accumulated 4π phase a physical 720° spatial rotation requires a separate bridge identifying θ^D progression with rotational orientation. PMM 3.7.6 does not currently supply that bridge.

The current event-level minimal A-record content is:

$$R^A_\alpha \;=\; \{\omega^A,\ a_{\mathrm{cl}},\ I^A\}$$

with endpoint instances $R^A_{i|\alpha}$ and $R^A_{j|\alpha}$. The event-level core does not explicitly store r_0, the sign of r_0, $\tilde{\Xi}^D$, or a rotation axis. The unnormalized carrier preserves the entry imbalance during D, but cross-FTU persistence therefore requires either a derivation showing that this information is already encoded in existing $A/\Phi/C_1/C_2$ content or a minimal additional record variable supplied through C_3.

No identification is made between $\tilde{\Xi}^D$ and δ_{ij}. The current theory retains:

δ_{ij} = independent primitive inter-participant distance

and the direct bridge between single-FTU D geometry and δ remains open.

Status: CANDIDATE conditional on coherent carrier persistence. The H_n sequence is algebraic once that condition is assumed. A spatial-rotation interpretation, full SU(2), and the persistence mechanism remain OPEN.

25.2 Angular Temporal Recursion Principle

Status: EXTENSION FIRST-PRINCIPLE POSTULATE. Inference from completed A records contributes to a generative B state for subsequent pairwise FTUs. Causal ordering governs which completed events can participate; angular history determines the possible relative phases. This adds a cross-FTU generative interpretation to the existing B → D → A → Φ → future B/D cycle.

The proposed grade bridge is:

$c^3(\alpha) \to$ A-record $\alpha \to \Phi \to$ angular history-dependent B preparation $\to c^0(\beta)$, with $\alpha \prec \beta$.

The relation $\alpha \prec \beta$ denotes causal precedence of completed α with respect to candidate β. Angular recurrence may revisit the same angle, while the causal event order distinguishes prior history from future events. No reverse traversal of an FTU is required.

Candidate B preparation map. Let $\Phi_{\prec\beta}$ contain only A records from the causal past of candidate event β, and let $\chi\beta$ represent the physical conditions of the receiving system, including any recording/playback apparatus. Propose:

$$\rho^B\beta \;=\; \mathcal{G}\theta(\Phi_{\prec\beta},\; \chi\beta)$$

Here $\rho^B\beta$ is a normalized positive B-state operator. The operator $\mathcal{G}\theta$ is an open physical law; its existence and form are the subject of the new postulate. The C_1-admissible state can then be conditioned using the existing $\Pi_a d_m$ projector:

$\rho^B_{\beta,\mathrm{adm}} = \Pi_a d_m \; \rho^B\beta \; \Pi_a d_m \;/\; Tr(\Pi_a d_m \; \rho^B\beta)$, provided the denominator is positive.

For an admissible candidate contact e:

$$P_e@ \;=\; Tr(\rho^B_{\beta,\mathrm{adm}}\; Q_e).$$

In the pure-state case, $\rho^B_{\beta,\mathrm{adm}} = |\Psi^B_a d_m\rangle\langle\Psi^B_a d_m|$, this reduces exactly to the current PMM expression $P_e@ = \langle\Psi^B_a d_m|Q_e|\Psi^B_a d_m\rangle$. This is a candidate generative extension of C_1, leaving its existing pairwise event definition intact.

Coherent interference branch — conditional mathematical test. If two causally prior records contribute coherent amplitudes at a candidate B channel, write:

$$u_1 \;=\; a_1 \;\exp(i\Theta_1/2)$$

$$u_2 \;=\; a_2 \;\exp(i\Theta_2/2)$$

$$I^B \;\propto\; |u_1 \;+\; u_2|^2 \;=\; a_1^2 \;+\; a_2^2 \;+\; 2a_1a_2 \;\cos[(\Theta_1 \;-\; \Theta_2)/2].$$

The half-angle correspondence is motivated by the existing candidate $U(\theta) = \exp[(\theta/2)\mathrm{J}]$ transport in §25.1. Θ represents accumulated cross-event transport angle; it is distinct from the local D angle θ^D, whose canonical closure is π. For $a_1 = a_2 = 1$, relative Θ differences of 0, 2π, and 4π give respective intensities 4, 0, and 4. This demonstrates mathematical compatibility with interference when a coherent phase-bearing cross-FTU channel is supplied.

A–B crossing interpretation (diagram correspondence). A crossing in the Multi-FTU Merge figure represents a candidate interference pairing between two distinct, causally prior A-history contributions to the same receiving B channel. If those two contributions are coherent, their unique unordered pair contributes one cross term $2a_ia_j\cos[(\Theta_i - \Theta_j)/2]$ to the B intensity. With N mutually coherent contributions to one B channel, the maximum number of distinct pairwise cross terms is $\mathrm{N}(\mathrm{N}-1)/2$. That count can be read directly from the drawn crossings only when each crossing denotes one unique contributing pair exactly once. Repeated line intersections, incoherent contributions, and

histories outside the causal past do not add independent interference terms. This is a diagram-to-equation convention for the conditional coherent branch, not a physical crossing law derived independently from $C_3/\Phi/C_1$.

Video recording/playback — causal consistency test. Original recording interactions create completed A records. A powered playback system accesses retained image information in an order chosen by the playback operation and generates fresh B preparation, D contacts, and A records. Playing three recorded images in order $1 \to 2 \to 3$, or in order $3 \to 2 \to 1$, creates new forward-progressing physical events in either case. The playback device supplies energy; the record supplies image information. Ordinary video may preserve image intensity information without preserving the original light's relative optical phase, so video playback uses the general ρ^B preparation branch. Coherent optical interference uses the additional phase-preserving branch.

Preservation test and scope boundary. The minimal A record $R^A = \{\omega^A,\, a_{\mathrm{cl}},\, I^A\}$ does not by itself reconstruct arbitrary relative interference phase. Retention of the original ψ^B, phase-bearing data, or a suitable cross-FTU coupling requires an explicit derivation through $C_3/\Phi/C_1$. The unnormalized carrier $\widetilde{\Xi}^D$ vanishes for $r_0 = 0$ and cannot alone represent every prepared B wave. Angular periodicity supplies neither causal selection nor energy for playback.

Two-test result: (1) video replay/reverse replay passes a causal consistency demonstration; (2) two coherent contributions pass the algebraic interference demonstration conditionally. Together these establish consistency of the proposed $\mathrm{A} \to \Phi \to \mathrm{B}$ connection with the two examples. A PMM-native physical generation law and independent experimental confirmation remain open. Snell-law recovery, gravity derivations, and broader reconstruction of existing physical theories lie outside these two tests.

25.3 A-Phase Information Capacity

The A-phase rail-area readout is the product of the two D-vantage scales:

$$\mathrm{F}^A_{tu} \;=\; \rho_1\rho_2 \;=\; (z^D)^2$$

The Planck-area reference and dimensionless capacity coordinate are:

$$A_{\mathrm{ref}} \;=\; 4\ell_p^2 \;=\; 4\hbar G_N/c^3$$

$$q_A \;=\; \mathrm{F}^A_{tu}/A_{\mathrm{ref}}.$$

For a horizon area A, the standard horizon-entropy normalization is $S_H/k_B = A/(4\ell_p^2)$. Expressing that entropy in bits gives $H_H = S_H/(k_B\ \ln 2)$. Applying the same area normalization to the A-phase rail area gives its maximum information-capacity readout:

$$H_{\max} \;=\; q_A/\ln 2 \;=\; \mathrm{F}^A_{tu}/(4\ell_p^2\ \ln 2).$$

Thus q_A counts horizon-normalized area units and $H_{\max}$ expresses the corresponding capacity in bits. The horizon normalization is an imported thermodynamic relation; its use for a general completed FTU is the A-phase capacity correspondence, not a separate derivation of the horizon entropy law.

26. A-Side Energy and Stress Record

A completed FTU carries an energetic readout:

$$E^A \;=\; \hbar\omega^A \;=\; E^D$$

with mass-equivalent readout:

$$m^A \;=\; E^A/c^2$$

The minimal mass-density contribution over relational volume V* is:

$$\rho^A \;=\; E^A/(c^2 V*)$$

A complete gravitational source can also carry the completed D event's response to deformation. The revised two-vantage geometry separates common scale, planar shape, and c-grade imbalance before any stress law is proposed.

26.1 Revised D Deformation Coordinates

The common spatial readouts are:

$$x^D \;=\; Le^{a}$$

$$y^D \;=\; Le^{-a}$$

$$z^D \;=\; L$$

Therefore the logarithmic spatial strains are:

$$\epsilon_x \;=\; d\ \ln\ x^D \;=\; d\ \ln\ L \;+\; da$$

$$\epsilon_y \;=\; d\ \ln\ y^D \;=\; d\ \ln\ L \;-\; da$$

$$\epsilon_z \;=\; d\ \ln\ z^D \;=\; d\ \ln\ L$$

so:

$$\epsilon_z \;=\; \tfrac{1}{2}(\epsilon_x \;+\; \epsilon_y)$$

The two c-grade vantage scales are:

$$\rho_1 \;=\; Le^{r}$$

$$\rho_2 \;=\; Le^{-r}$$

with differential responses:

$$d\ \ln\ \rho_1 \;=\; d\ \ln\ L \;+\; dr$$

$$d\ \ln\ \rho_2 \;=\; d\ \ln\ L \;-\; dr$$

Thus a and r carry different deformation information:

$$a \;=\; \text{planar } x\text{-}y \text{ shape}$$

$$r \;=\; c^1\text{-}c^2 \text{ vantage imbalance}$$

At anti-contact, $r \to 0$; planar shape a is unconstrained by the scale lock.

26.2 Native D Energy Functional — Open

No active candidate D interaction-energy law is adopted in this release. The native interaction energy remains an open derivation target. Its most general current dependence can be written schematically as:

$$E^D \;=\; E^D(L,\; a,\; r,\; \theta^D,\; \text{carrier data},\; ...)$$

Any future candidate must be specified independently and then tested against the existing PMM geometry. It must admit the massive and null sectors and support acceptable stress behavior without defining an auxiliary variable from the same equation being tested.

26.3 Stress Readout

Directional stress remains a downstream readout of an independently supplied D interaction energy functional:

$$p_u \;=\; -(1/V)\;\partial E^D/\partial \epsilon_u$$

with u ∈ {x,y,z} and spatial strain channels ϵ_x, ϵ_y, and ϵ_z. Since $\epsilon_z = \frac{1}{2}(\epsilon_x + \epsilon_y)$, these partial derivatives require a specified extension of E^D to independent strain channels. Without such an extension, only responses to independent variations of ln L and a are defined; separate directional stresses remain open.

The c-grade imbalance r can also produce an internal vantage-response channel once a native E^D is known:

$$I_r^D \;=\; -(1/V)\;\partial E^D/\partial r$$

I_r^D is an internal D response coordinate. Its bridge to ordinary spatial stress is OPEN.

At closure, C_3 transcribes the completed energy-and-response content into the A record. Let $\rho_{E,\alpha}$ denote the local energy density and $V*_\alpha$ its associated event-record volume. In a common local rest-frame recovery representation, define the extensive record contribution:

$$\boldsymbol{\mathcal{R}}^A_{\mu\nu,\alpha} \;\to\; V*_\alpha\;\text{diag}(\rho_{E,\alpha},\; p_1\alpha,\; p_2\alpha,\; p_3\alpha)$$

and many completed FTUs can build an effective stress-energy source:

$$T_{\mu\nu} \;=\; (1/V*)\;\sum\;\text{distinct completed }\alpha\;\boldsymbol{\mathcal{R}}^A_{\mu\nu,\alpha}$$

Here V* is the averaging-region volume; each $\boldsymbol{\mathcal{R}}^A_{\mu\nu,\alpha}$ has energy units, so $T_{\mu\nu}$ has energy-density units. Source deduplication and the physical record-volume assignment remain open.

26.4 Status

Current status:

two-vantage D deformation geometry

→ active

$$\epsilon_z \;=\; \tfrac{1}{2}(\epsilon_x \;+\; \epsilon_y)$$

$$\to\;\text{derived from } x^D \;=\; Le^a,\; y^D \;=\; Le^{-a},\; z^D \;=\; L$$

native E^D interaction functional

$$\to\;\text{OPEN}$$

spatial stress and internal vantage response

→ downstream of a future independently specified E^D functional

The principal remaining derivation target in this sector is therefore a native D interaction energy functional compatible with the revised two-vantage geometry.

27. Multi-FTU Gravity Lineage

The gravity sector begins at the multi-FTU vantage and uses relational clock variables rather than c-order labels as direct dynamical inputs.

Its canonical lineage is:

completed FTU

→ two endpoint record instances $R^A_{i|\alpha}$, $R^A_{j|\alpha}$ / one event identity α

$$\rightarrow \text{ accumulated lineage histories } \Phi$$

→ active-source extraction $\mathcal{S}[\Phi] = s^A$

$$\rightarrow \text{ Clock Flux on the } \delta \text{ network determines } \varphi$$

→ $\kappa = e^{-\varphi}$ gives the gravity-sector lineage clock response

→ C_2 supplies the general path constraint for later D realization

The single-FTU c-order architecture defines what an FTU is. The multi-FTU gravity architecture describes how completed FTUs influence later D-phase timing.

Every step is tagged below as one of:

CORE — already present in the PMM architecture

DEFINITION — introduces notation and readouts

DERIVED — follows from prior stated assumptions

POSTULATE — new physical content not forced by prior mathematics

CANDIDATE — proposed relation not yet derived

RECOVERY — comparison or normalization against established physics

OPEN — active derivation target

Compound labels combine these statuses when a statement has more than one role.

This status discipline is part of the canonical theory. Recovery status leaves upstream inputs classified according to their native PMM role.

28. Native D Clock Response

28.1 Core inputs

CORE

PMM states:

Time is angular.

$$\mathrm{F}^T_{tu} \;=\; ([\hbar/(m_p c^2)]\Delta\theta^B,\; d/c,\; \Delta\theta^A/\omega^A)$$

$$\mathrm{F}^G_{tu} \;=\; \nabla\varphi$$

$$g \;=\; c^2 \mathrm{F}^G_{tu}$$

$$C_2 \;=\; C_2[\Phi, \delta, \Sigma]$$

C_2 is the general D-path constraint. Clock Flux determines the gravity-sector κ. The explicit gravity-sector response is:

$$\Phi \;\rightarrow\; \mathcal{S}[\Phi] \;=\; s^A \;\rightarrow\; \text{Clock Flux on } \delta \;\rightarrow\; \varphi \;\rightarrow\; \kappa \;=\; e^{-\varphi}.$$

Any gravitational role for Σ must enter through a separately derived contribution to $\mathcal{S}[\Phi]$ or another explicit bridge; no direct $\Sigma \rightarrow \kappa$ law is assumed here.

The gravitational variable uses a dimensionless comparison of D angular rates sampled on participant lineages. Event angular progression belongs to each incident FTU; the normalized clock response belongs to the lineage node. This relational multi-FTU construction uses lineage-local normalized D-clock behavior as its direct dynamical input.

28.2 Dimensionless angular-rate response

DEFINITION

Let event α be incident on participant lineage $\mathcal{P}_i$. Let t denote the coordinate / relational comparison time already used in the D c^1 readout. Define the lineage D angular-rate response by:

$$\kappa(i|\alpha) \;=\; [(d\theta^D/dt)(i|\alpha)] \;/\; [(d\theta^D/dt)_0(i|\alpha)].$$

The normalization removes the intrinsic clock scale and compares how rapidly D phase progresses relative to the common comparison time. The normalized clock response therefore belongs naturally to the participant lineage rather than to FTU α as an object.

POSTULATE — Lineage Clock State

At relational slice n, each participant lineage $\mathcal{P}_i$ carries one local clock state κ_i^n. Every concurrent incident FTU that samples that lineage in the same slice uses that same local state:

$$\kappa(i|\alpha) \;=\; \kappa_i^n.$$

The Lineage Clock State Postulate assigns one κ_i^n to all incident FTUs sampling lineage i in slice n.

Interpretation:

$$\kappa_i \;=\; 1 \;\rightarrow\; \text{neutral } D \text{ progression}$$

$$\kappa_i \;<\; 1 \;\rightarrow\; \text{slowed } D \text{ progression}$$

$$\kappa_i \;>\; 1 \;\rightarrow\; \text{accelerated } D \text{ progression.}$$

On the conditional massive/timelike action branch, intrinsic $T^D = \tau$. κ compares D phase progression with respect to the comparison time t. For identical intrinsic massive clocks:

$$(d\theta^D/dt) \;=\; (d\theta^D/d\tau)(d\tau/dt)$$

so, when the intrinsic $d\theta^D/d\tau$ factor is the same in the local and neutral reference states:

$$\kappa \;=\; (d\tau/dt)/(d\tau/dt)_0.$$

Because κ is defined from angular progression with respect to the comparison time rather than from proper time itself, a null carrier has:

$$d\tau^D \;=\; 0$$

but

$$\Delta\theta^D \;\neq\; 0.$$

The angular-rate definition therefore preserves a lineage clock-state readout in the null sector without assigning proper time to the photon.

29. Relational Clock Potential and Gravity

29.1 Common-rate invariance

A gravitational relation is invariant when every local clock response is rescaled by the same positive factor:

$$\kappa_i \;\to\; s\kappa_i$$

$$\kappa_j \;\to\; s\kappa_j$$

Edge observables depend on relative clock behavior.

DEFINITION

Introduce the dimensionless relational clock potential:

$$\varphi_i \;=\; -\ln\,\kappa_i$$

Using any incident event α sampling the same lineage clock state:

$$\varphi_i \;=\; -\ln\{[(d\theta^D/dt)(i|\alpha)]/[(d\theta^D/dt)_0(i|\alpha)]\}$$

For the identical massive-clock branch this is equivalently:

$$\varphi_i \;=\; -\ln\{[(d\tau/dt)_i]/[(d\tau/dt)_0]\}.$$

Then a common rescaling shifts every φ by the same constant and therefore cancels from all differences.

For relational edge $i \leftrightarrow j$:

$$\Gamma_{ij} \;=\; (\varphi_j \;-\; \varphi_i)/\delta_{ij}$$

Γ_{ij} is the dimensionless D-clock gradient per relational length.

29.2 Gravitational acceleration readout

DEFINITION / CORE-MOTIVATED READOUT

The D crease c converts the inverse-length clock gradient to acceleration:

$$g_{ij} \;=\; c^2\ \Gamma_{ij}$$

Continuum form:

$$g \;=\; c^2 \nabla \varphi$$

$$\text{Equivalently, since } \varphi \;=\; -\ln \kappa :$$

$$g \;=\; -c^2 \nabla \ln \kappa$$

Using the normalized angular-rate definition:

$$g \;=\; -c^2 \nabla \ln[(d\theta^D/dt)/(d\theta^D/dt)_0].$$

For the identical massive-clock branch:

$$g \;=\; -c^2 \nabla \ln[(d\tau/dt)/(d\tau/dt)_0].$$

Thus g points toward decreasing normalized proper-time rate, the locally slower clock side, consistent with the attractive sign convention below.

The normalized FTU gravity readout is therefore:

$$\mathrm{F}^G_{tu} \;=\; \nabla\varphi \;=\; -\nabla \ln[(d\theta^D/dt)/(d\theta^D/dt)_0]$$

with:

$$g \;=\; c^2 \mathrm{F}^G_{tu}$$

Gravity is therefore the relational gradient of normalized D-clock response. This keeps the human-facing D-time interpretation while making the dimensions explicit.

29.3 Sign convention

A positive completed source slows local D progression:

source present

$$\rightarrow\ \kappa \;<\; 1$$

$$\rightarrow\ \varphi \;>\; 0$$

For an isolated attractive source, φ decreases outward. Therefore:

$$\nabla\varphi \text{ points inward}$$

and:

$$g \;=\; c^2 \nabla \varphi$$

has the correct attractive orientation.

30. How Φ Conditions the D Clock

The multi-FTU source-response chain is:

completed endpoint A records → lineage Φ histories → event-ID-deduplicated active-source extraction → Clock Flux on $\delta \to \varphi \to \kappa$.

The network-response shorthand:

$$\kappa_i \;=\; \mathrm{K}(\Phi_i,\ \text{local } \delta\text{-relations},\ \ldots)$$

is not treated as an additional independent local constitutive law.

Let the active completed-record source extraction be:

$$s^A \;=\; \mathcal{S}[\Phi].$$

The native extraction map $\mathcal{S}$ is OPEN. It must distinguish persistent history from currently active source content and must avoid double counting shared-event references.

Once s^A is supplied, the discrete Clock Flux law of §33 determines the relational clock potential φ on the δ network, subject to the common-rate gauge or physical boundary data. The clock response is then:

$$\kappa_i \;=\; e^{-\varphi_i}.$$

K is the **derived network solution functional**:

$$\Phi \;\to\; \mathcal{S}[\Phi] \;=\; s^A$$

$$s^A \;+\; \delta \;\to\; \text{Clock Flux} \;\to\; \varphi$$

$$\varphi \;\to\; \kappa \;=\; e^{-\varphi}.$$

Status: the direct clock-response blocker is CLOSED at the architectural level. The active source-extraction map $\mathcal{S}[\Phi]$ remains OPEN / CANDIDATE.

TESTABLE GRAVITY CLAIM

Given independently specified s^A, δ, λ^G, and boundary/gauge data, Clock Flux predicts φ_i and therefore $\kappa_{i,\mathrm{pred}} = e^{-\varphi i}$. The physical claim is that a measured lineage clock response $\kappa_{i,\mathrm{meas}}$ matches $\kappa_{i,\mathrm{pred}}$ within the regime where the PMM gravity model applies. A reproducible mismatch under fixed source and boundary reconstruction falsifies this gravity-sector mapping. The identity $\kappa = e^{-\varphi}$ itself remains definitional because $\varphi = -\ln \kappa$.

30.1 Reduced additive-source consequence

If, in a reduced regime, active source extraction is additive:

$$s^A \;=\; s_1^A \;+\; s_2^A,$$

then the linear Clock Flux equation gives:

$$\varphi_{12} \;=\; \varphi_1 \;+\; \varphi_2$$

under one fixed gauge/boundary convention.

Because:

$$\kappa \;=\; e^{-\varphi},$$

it follows that:

$$\kappa_{12} \;=\; \kappa_1\kappa_2.$$

The multiplicative response follows in an additive-source regime from source addition, linear Clock Flux in φ, and $\varphi = -\ln\kappa$.

A global proportionality $\varphi \propto \Phi_s$ is **not** implied unless the eventual source-extraction map $\mathcal{S}$ supplies that additional relation.

30.2 Current-source reconstruction — candidate research decomposition

CANDIDATE / OPEN — not part of the canonical gravity foundation

Persistent history Φ is not itself current source mass. A candidate bookkeeping decomposition is:

$\Phi \to$ lineage-frontier selection $\to$ event-ID deduplication $\to$ current carried lineage source $\mu^n \to$ placement on the δ network $\to s^A$.

For the candidate source decomposition, define a history frontier $\mathfrak{F}_i^n$ for lineage i as the maximal completed records under the lineage order generated by $\sim^L$. This prevents an old completed event from being counted again merely because it remains in Φ, while event identifiers prevent the two participant references to one shared FTU from being counted as two independent energies.

The frontier rule does not by itself determine the present source carried by a lineage. Each endpoint record of event α references the same one event-level energy $E^A(\alpha)$; the second record is not a second source-energy copy. The internal D-vantage labels 1 and 2 are not permanent participant identities. Therefore the conditional seed weights $p_1|_e@$ and $p_2|_e@$ cannot be used to partition E^A between participant lineages.

A closed pairwise source-state update requires a signed event-to-lineage transfer law:

$$\Delta E_i|\alpha \;+\; \Delta E_j|\alpha \;=\; 0$$

with:

$$\mu_i^{n+1} \;=\; \mu_i^n \;+\; \Delta E_i|\alpha/c^2,$$

$$\mu_j^{n+1} \;=\; \mu_j^n \;+\; \Delta E_j|\alpha/c^2.$$

The derivation of $\Delta E_i|\alpha$ is OPEN in the native D interaction / C_2 outcome problem of §26 and the Recovery companion. Current source uses active carried energy extracted from history.

31. Record Influence Continuity

C_3 already provides persistence:

C_3 = what persists and is transcribed into Φ

and the multi-FTU history update accumulates completed endpoint records:

$$\Phi_i^{n+1} \;=\; \mathrm{Append}(\Phi_i^n,\; R^A_{i|\alpha(i,n)})$$

R^A_α is the event-level minimal numerical record (§15). The update appends its lineage-endpoint instance $R^A_{i|\alpha}$. Any additional retained information requires a specified transcription and accumulation rule.

Record Influence Continuity supplies the spatial flux principle.

31.1 Gravity-sector postulate

POSTULATE — Record Influence Continuity

> Completed A endpoint records condition later D realization through the relational network. For one shared event, their common source content is counted once by event identity. Away from completed-record sources, clock influence is locally conserved.

It contains two claims:

locality:

influence propagates through FTU relations

continuity:

source-free relational regions conserve clock influence

This postulate defines the relational continuity principle upstream of the continuum field equation.

32. Clock Flux

32.1 Source-free consequence

Given Record Influence Continuity, a source-free relational region has zero net production of the clock-gradient field.

DERIVED FROM THE POSTULATE

In continuum notation:

$$\nabla \cdot (\nabla \varphi) \;=\; 0$$

or:

$$\nabla^2 \varphi \;=\; 0$$

in a source-free region after the continuum limit is valid.

The discrete form is primary. The continuum equation is the candidate large-network representation, pending derivation of the many-FTU coarse-graining limit.

32.2 Enclosed-source form

For a closed relational region V^R, continuity requires the outward boundary flux to equal the negative enclosed completed-record source multiplied by one coupling normalization:

$$\oint \partial V^R \; \nabla \varphi \;\cdot\; dA \;=\; -\lambda_G \; Q^A(V^R)$$

where:

$Q^A(V^R)$ = total completed gravitational record-source inside V^R

and:

$$[\lambda_G] \;=\; \text{length / mass}$$

$$if\ Q^A \text{ is represented as } a \text{ mass-equivalent source.}$$

The negative sign is fixed by the convention that positive source gives $\varphi > 0$ near the source while outward flux is negative.

This is the **Clock Flux Law**.

Status:

Record Influence Continuity — POSTULATE

Clock Flux Law — DERIVED from continuity + locality + additive enclosed source

coupling normalization λ_G — OPEN until fixed by recovery or deeper derivation

32.3 Minimal A-side source candidate

The single-FTU readouts already provide:

$$E^A \;=\; E^D$$

$$m^A \;=\; E^A/c^2$$

The minimal scalar source candidate is the formal sum below, with shared-event source energy counted once by event identity. Active-source extraction must distinguish successive history records from current carried mass; that extraction is OPEN:

$$Q^A \;=\; \sum \text{ completed } m^A$$

$$= \sum \text{ completed } E^A/c^2$$

CANDIDATE

This is the lowest-order source identification. The full relativistic source can also include pressure, anisotropic stress, and additional completed-record structure.

33. Discrete δ-Network Clock Flux

The multi-FTU theory is fundamentally relational, so the discrete law precedes the continuum law.

For each relational edge $i \leftrightarrow j$:

$$\Gamma_{ij} \;=\; (\varphi_j \;-\; \varphi_i)/\delta_{ij}$$

Let $A*_{ij}$ denote the dual relational face area associated with boundary edge $i \leftrightarrow j$ in the assembled tetrahedral network.

For relational region V^R:

$$\sum_{i\leftrightarrow j \,\in\, \partial V^R} A*_{ij}\ \Gamma_{ij}$$

$$= -\lambda_G \, Q^A(V^R)$$

This is the discrete Clock Flux equation.

For an interior node i, the corresponding discrete divergence form is:

$$\sum_j [A *_{ij} / \delta_{ij}] \, (\varphi_j - \varphi_i)$$

$$= -\lambda_G \, s_i^A$$

where s_i^A is the completed record-source assigned to the node or its dual cell.

Status:

$$\delta_{ij} \text{ — existing multi-FTU primitive}$$

$A*_{ij}$ — derived from the chosen relational cell complex

φ — relational D-clock potential

source-free discrete balance — derived from Record Influence Continuity

$$\text{source coupling } \lambda_G \text{ — OPEN}$$

No global spatial direction field is required.

33.1 Clock-response solution functional

Define the positive weighted relational graph Laplacian:

$$(\mathcal{L}\delta\varphi)_i = \sum_j [A *_{ij} / \delta_{ij}](\varphi_i - \varphi_j).$$

Then the discrete Clock Flux equation is:

$$\mathcal{L}\delta\varphi = \lambda_G s^A.$$

With active source extraction:

$$s^A = \mathcal{S}[\Phi],$$

the solution is fixed by the δ-network and source after imposing physical boundary data or fixing the common additive gauge. Formally:

$$\varphi = \lambda_G \mathcal{L}\delta^+ \mathcal{S}[\Phi]$$

where $\mathcal{L}\delta^+$ denotes the inverse on the fixed-boundary subspace or the gauge-reduced pseudoinverse on a connected network.

Therefore:

$$\kappa_i = \exp[-\lambda_G(\mathcal{L}\delta^+ \mathcal{S}[\Phi])_i].$$

The solution functional K is:

$$\mathrm{K}[\Phi, \delta] \;=\; \exp[-\lambda_G \mathcal{L} \delta^{+} \mathcal{S}[\Phi]]$$

up to the common-rate gauge or specified boundary condition.

A common shift $\varphi_i \to \varphi_i + \mathrm{C}$ rescales every κ_i by the same positive factor and leaves Γ_{ij} and g_{ij} invariant. In a source-free connected closed network, $\mathcal{L}\delta\varphi = 0$ gives constant φ modulo that gauge; with nontrivial boundary data, φ is relationally harmonic.

Status: DERIVED from the discrete Clock Flux law once s^A, δ, λ_G, and boundary/gauge data are supplied. The unresolved native physics lies in $\mathcal{S}[\Phi]$ and, if sought, λ_G beyond recovery.

34. Consistency Checks

The following checks test the gravity chain before downstream GR recovery.

34.1 Dimensional consistency

$$\varphi \text{ dimensionless}$$

$$\nabla\varphi \; L^{-1}$$

$$c^2 \nabla\varphi \; L \; T^{-2}$$

so:

$$g \;=\; c^2 \nabla\varphi$$

has acceleration dimensions.

For the flux law:

$$\oint \nabla\varphi \cdot dA \; L$$

therefore a mass-equivalent source requires:

$$[\lambda_G] \;=\; L/M$$

and:

$$[G_N/c^2] \;=\; L/M$$

provides the accepted gravitational crease scale.

34.2 Common-rate invariance

Under:

$$\kappa_i \;\to\; s\kappa_i$$

we have:

$$\varphi_i \;\to\; \varphi_i \;-\; \ln\, s$$

so edge differences and gravity are unchanged.

34.3 Massive-sector kinematic consistency

Massive FTUs have nonzero D progression and therefore admit κ directly.

The massive sector is kinematically consistent.

34.4 Null-sector kinematic consistency

Null carriers satisfy:

$$d\tau^D \;=\; 0$$

and carry angular D progression:

$$d\theta^D \;\neq\; 0$$

Because κ is an angular-rate ratio, the gravitational clock variable is defined for the null sector.

34.5 A-record recursion

The source lineage is:

D completion

$\rightarrow$ C_3 endpoint transcription

$$\rightarrow\; R^A_{i|\alpha},\; R^A_{j|\alpha}$$

$$\rightarrow\; \text{lineage } \Phi \text{ histories}$$

$$\rightarrow\; \text{later } C_2$$

$$\rightarrow\; \kappa$$

so gravity is downstream of completed record persistence.

34.6 Attractive sign convention

Positive source:

$$\kappa \;<\; 1$$

$$\rightarrow\; \varphi \;>\; 0 \text{ near source}$$

$$\rightarrow\; \varphi \text{ decreases outward}$$

$$\rightarrow\; \nabla\varphi \text{ points inward}$$

$$\rightarrow\; g \;=\; c^2\nabla\varphi \text{ points inward}$$

The sign is internally consistent.

34.7 Newtonian normalization — RECOVERY

RECOVERY

Newtonian normalization is detailed in the separate recovery documentation, Recovery R1.

The recovery uses the PMM gravity chain and candidate minimal source $Q^A = \mathrm{M}$, then tests the weak-field Newtonian normalization. Recovery results remain downstream of the canonical PMM foundation.

34.8 Foundational target

The foundational gravity target is:

Derive the full completed-record source Q^A from PMM's event architecture and determine the native status of λ_G relative to the accepted gravitational crease G_N.

34.9 Temporal-rank recovery candidates

CANDIDATE / RECOVERY — not first principles

The temporal geometry motivates downstream recovery candidates that are tested in the Recovery companion rather than adopted as native PMM laws. The gravity clock is a normalized D angular rate with respect to comparison time t:

$$\kappa \;=\; (d\theta^D/dt)/(d\theta^D/dt)_0.$$

κ is defined with respect to comparison time t. A radial length relation requires the independent static spherical bridge $d_0 = \mathrm{dr}$ and $\mathrm{d} = d\ell_r$. Under that bridge, the candidate Clock–Distance Reciprocity relation is:

$$\mathbf{d}\ell_{\mathbf{r}} \;=\; \mathbf{dr}/\boldsymbol{\kappa}$$

and therefore:

$$h_{rr} \;=\; \kappa^{-2}.$$

This identification is a recovery bridge. The core still leaves the physical relation between the single-FTU length readout d and multi-FTU inter-participant distance δ open.

A candidate static covariant completion of Clock Flux uses:

$$\Box\varphi \;=\; 0$$

with:

$$\varphi \;=\; -\ln\kappa.$$

For a static metric $ds^2 = -\kappa^2 c^2 dt^2 + h_{ij}dx^i dx^j$, this is equivalent to:

$$\Delta_h \kappa \;=\; 0.$$

Combined with Clock–Distance Reciprocity in static spherical symmetry, this recovery route yields the Schwarzschild exterior lapse and radial metric. A broader candidate clock–curvature functional is:

$$\boldsymbol{\mathcal{I}}_{\mathrm{stat}} \;=\; C^G \int \kappa\, R[h]\, \sqrt{h}\; d^3x$$

with a relational discretization of Regge type schematically proportional to:

$$\sum_e \bar{\kappa}_e \ \delta_e \ \Delta_e.$$

These relations remain recovery candidates until derived from the native δ-network dynamics and the PMM event/record architecture.

34.10 Static-geometry bridge audit — synchronized recovery status

CANDIDATE / RECOVERY / REJECTED-BRANCH AUDIT — not first principles

The static-gravity audit distinguishes a candidate edge bridge, a spherical recovery branch, a rejected universal ansatz, and an open nonlinear target.

Candidate edge bridge. For a realized FTU α with $e@(\alpha) = \{\mathrm{i,j}\}$, test:

$$\mathbf{d}\boldsymbol{\alpha} \ \leftrightarrow \ \boldsymbol{\delta}^{\mathbf{n}}_{\mathbf{ij}}$$

at the relational slice in which the pair relation is realized. This identifies the D traversal length entering $T^D = \mathrm{d/c}$ with the inter-lineage edge length used by the δ network. It does not identify the internal single-FTU depth z^D directly with δ_{ij}, and it remains an open PMM-native derivation target.

Spherical areal-radius branch. In spherical symmetry the transverse geometry defines areal radius by:

$$A \ = \ 4\pi r^2.$$

The recovery identification $d\ell_r = dr/\kappa$ then yields $h_{rr} = \kappa^{-2}$ and, together with the candidate covariant clock equation, reproduces the Schwarzschild exterior. This is a conditional spherical recovery rather than a universal directional law.

Rejected universal longitudinal-only ansatz. Let $\bar{h}_{ij}$ be a flat reference metric and test:

$$h_{ij} \ = \ \bar{h}_{ij} \ + \ (\kappa^{-2} \ - \ 1) n_i n_j,$$

$$n \parallel \nabla\kappa,$$

while keeping the transverse metric unchanged. In the weak static vacuum test $\kappa = 1 + \mathrm{u}$ with $\mathrm{u} = \epsilon xy$ and $\nabla^2 u = 0$, the static-vacuum target requires $R_{xy} = \partial_x \partial_y u = \epsilon$ at first order, whereas this ansatz gives $R_{xy} = 0$. At x = y it also gives $R_{xx} = -\epsilon$ while the target has $\partial_x^2 u = 0$. The longitudinal-only ansatz is therefore rejected as a general static bridge.

Native D common-scale branch. The existing D geometry has:

$$x^D \ = \ Le^a,$$

$$y^D \ = \ Le^{-a},$$

$$z^D \ = \ L.$$

Thus a change of the common scale L necessarily changes the mean x-y scale together with z^D. Under the recovery candidate:

$$L/L_0 \ = \ \kappa^{-1} \ = \ e^{\varphi},$$

with a = 0 and fixed local principal orientation on a flat reference chart, the spatial metric becomes:

$$h_{ij} \;=\; e^{2\varphi}\;\bar{h}_{ij}.$$

For $\kappa = e^{-\varphi}$ and source-free flat-reference $\Delta\varphi = 0$, this ansatz satisfies the general static-vacuum tensor equation $R_{ij} = \kappa^{-1} D_i D_j \kappa$ at first order for an arbitrary weak harmonic φ. The exact residual in this restricted common-scale ansatz is:

$$R_{ij} \;-\; \kappa^{-1} D_i D_j \kappa \;=\; -2(\partial_i\varphi)(\partial_j\varphi) \;-\; \bar{h}_{ij}\;\Delta\varphi,$$

so in the source-free flat-reference branch:

$$\mathbf{R_{ij}} \;-\; \boldsymbol{\kappa^{-1}} \mathbf{D_i D_j} \boldsymbol{\kappa} \;=\; -\mathbf{2}(\boldsymbol{\partial_i \varphi})(\boldsymbol{\partial_j \varphi}).$$

The common-scale branch therefore passes general static recovery at linear order and fails by a quadratic tensor unless additional transverse/shear dynamics contribute.

Transverse representation capacity. In a local principal frame the two-dimensional transverse metric can be parameterized as:

$$q_\perp \;=\; L^2\; R(\psi)^T\; \mathrm{diag}(e^{2a}, e^{-2a})\; R(\psi),$$

with $\det q_\perp = L^4$. The native variables L and a supply transverse area scale and anisotropy. The orientation ψ, its cross-FTU transport, and the dynamical laws for a and ψ are not derived. This decomposition shows that the present D variables can carry the required transverse geometry; it does not derive the gravitational response.

The decisive native target is therefore no longer another Schwarzschild fit. It is the full M_{op} evolution law coupling the δ geometry to the D transverse variables L, a, and orientation under the κ / φ gravity response, followed by comparison with the general static-vacuum equations.

35. Single-FTU / Multi-FTU Coupling

PMM presents the two sectors separately and couples them dynamically.

Single FTU supplies:

$B \to D \to A$ event geometry

D angular progression

one event-level $E^A(\alpha)$

two endpoint A-record instances carrying R^A_α

C_3 persistence

Multi-FTU supplies:

δ network

accumulated history Φ

relational D-clock response κ

clock potential φ

clock-gradient field

future C_1/C_2 conditions

The feedback cycle is:

completed FTU

$$\downarrow$$

two endpoint A records / one event identity α

$$\downarrow$$

accumulated lineage histories Φ

$$\downarrow$$

active-source extraction $\mathcal{S}[\Phi] = s^A$

$$\downarrow$$

Clock Flux on the δ network

$$\downarrow$$

φ = relational clock potential

$$\downarrow$$

$\kappa = e^{-\varphi}$ = gravity-sector D angular-rate response

$$\downarrow$$

C_2-constrained later D progression

$$\downarrow$$

future FTU realization

This defines one recursive theory with two vantages:

$$\text{single FTU} \;=\; \text{creases} \;+\; c\text{-order} \;+\; \text{action}$$

$$\text{multi-FTU} \;=\; \Phi \;+\; \text{timing response} \;+\; \text{gravity}$$

36. Theory Status

Single-FTU vantage — foundation

Core structure: creases + c-order + action

B/D/A event architecture

@ and $-$@

c-order mapping

two x-y D vantages

relational depth from two vantages

L/r/a decomposition

derived symmetric / antisymmetric D-vantage readouts ρ_+^D, ρ_-^D with invariant $(\rho_+^D)^2 - (\rho_-^D)^2 = (z^D)^2$

candidate physical interpretation of V_1 / V_2 as reciprocal null/light-channel vantages of one D interaction

closure spread Σ

$$\theta_{\mathrm{cl}} \;=\; \pi$$

minimal harmonic closure

space window

massive sector

null sector

A transcription

energy continuity

FTU readout family

phase-action ruler with $dS^D = \hbar d\theta^D$ and variable B/A spans

$\mathrm{F}_{tu}^T = (T^B, T^D, T^A)$ as the physical temporal coordinate map; independent-d and active conditional d = L parameterizations are rank three, while completed conditional d = L closure has rank at most two; orthogonal temporal-cube metric not assumed

candidate B span $\Delta\theta^B = \arccos\sqrt{P_e}$@ from the C_1-selected pair-contact channel

A information span $\Delta\theta^A = I^A = -\ln P_e$@ $= \ln \Omega_A$ with $\Omega_A = 1/P_e$@

completed action profile $\Delta\theta^B \to \pi \to \pi \to \Delta\theta^A$ across the c-order readout

outwardness / inwardness

derived unnormalized half-angle carrier $\widetilde{\Xi}^D = (r, -2r')^T$ of the P5 harmonic law; physical spinor interpretation remains candidate

pairwise realized contact: each elementary physically actualized pairwise interaction is one FTU; each actualized binary pair-contact proposition produces one unordered pair e@ for elementary D realization

concurrent event accounting: causally unordered compatible pair contacts may coexist as distinct FTU event identifiers in one relational slice, including contacts that share one participant lineage; they do not merge into a many-participant FTU

$\sum P_e$@ $= 1$ applies to complete mutually exclusive C_1 contact blocks; compatible binary pair contacts may actualize concurrently

pair-selection probability P_e@ $= P_{ij}$@ $= P_{ji}$@ is distinct from the conditional D-vantage seed weights $p_1|_e$@, $p_2|_e$@

r_0 and $\tilde{\Xi}^D$ signs refer to the chosen D-vantage order, not intrinsic participant identity or polarity

Single-FTU refinement:

derive or reject native C_3 selection of $A_{\mathrm{cl}} = A_G$ at the gravitational-crease limiting boundary

Action Cube candidate readout: F^{s}_{tu} denotes the Action Cube, $\mathrm{F}^{\mathrm{led}}_{tu}$ retains the additive ledger, and F^{sv}_{tu} retains the coordinate tuple (§4.1.1). The cube is not a new independent parameter, closure count, or adopted physical volume law.

Multi-FTU vantage — relational scaffold

Core structure: Φ + timing response + gravity

CORE

M_{op} — Motion Operations Principle: space is emergent relational motion; δ is realized spatial relation, evolution of δ is relational motion, second relational change is acceleration/deceleration, and directionality emerges from the relational geometry

$$\delta_{ij} \text{ as independent inter-participant distance}$$

$$A \to \Phi \to \text{future-event recursion}$$

Angular Temporal Recursion Principle (extension postulate): prior A records generate later B possibilities via Φ; coherent histories admit interference

$$\mathrm{F}^{G}_{tu} \;=\; \nabla\varphi \;=\; -\nabla \ln[(d\theta^D/dt)/(d\theta^D/dt)_0]$$

$$g \;=\; c^2 \mathrm{F}^{G}_{tu}$$

$$C_2[\Phi, \delta, \Sigma]$$

DEFINITION / CORE-MOTIVATED READOUT

$$\kappa \;=\; D \text{ angular-rate response}$$

$$\varphi \;=\; -\ln \kappa$$

$$\Gamma_{ij} \;=\; \Delta\varphi/\delta_{ij}$$

$$g \;=\; c^2 \nabla\varphi$$

POSTULATE

Pairwise Realization Principle (extension postulate): elementary realized contact actualizes one unordered pair e@

Lineage Clock State: each lineage carries one single-valued κ^n_i per relational slice, sampled by all concurrent incident FTUs

Record Influence Continuity

DERIVED FROM THE POSTULATE

source-free clock-flux conservation

closed-region Clock Flux form

discrete δ-network flux equation

CANDIDATE

$\mathcal{G}\theta(\Phi_{\prec\beta}, \chi\beta)$ as the candidate generative B-preparation map; pure-state contact probability is recovered as a special case

$C_1[\Phi] \to \mathcal{H}_e^B[\Phi]$ as the native contact-admissibility map; $Q_e = \mathrm{Proj}(\mathcal{H}_e^B[\Phi])$ as its sharp-projector representation

internal reciprocal decomposition $R_{e1} + R_{e2} = Q_e$ and derived D seed weights from the selected B state

$$Q^A \;=\; \sum E^A/c^2 \text{ as minimal scalar source candidate}$$

$\kappa_{12} = \kappa_1\kappa_2$ as a derived consequence in an additive-source Clock Flux regime

$H_n = [U(\pi)]^n$ as a relational half-angle transport sequence, conditional on coherent carrier persistence across completed FTUs $d\alpha \leftrightarrow \delta_{ij}$ as the candidate event-edge bridge for a realized pair FTU; spherical Clock–Distance Reciprocity remains recovery-only

OPEN

native D interaction energy functional of §26

native dynamical derivation of the Lineage Continuity relation $\sim^L$ beyond its postulated role

native active source-extraction map $\mathcal{S}[\Phi]$ from persistent record history to current carried source s^A, including the event-to-lineage energy-transfer / μ update law

additional noncommuting generators and any full SU(2) transport from PMM-native geometry

RECOVERY

Newtonian normalization → Recovery R1 (separate recovery documentation)

Canonical gravity foundation boundary

The canonical gravity foundation consists of the relational δ geometry, M_{op} relational-motion principle, completed-record history Φ, D-phase clock response, and the normalized D-clock-gradient readout $\mathrm{F}_{tu}^G = \nabla\varphi$. The definitions and asserted principles in this document specify the gravity foundation.

Recovery Summary

Detailed recovery derivations and prediction tests are maintained in separate companion research documents.

The recovery program connects PMM's BDA event architecture to established physical descriptions through explicitly identified limiting relations. The index identifies each construction, its mathematical basis, classification, and reference. Born weighting and A-phase capacity are indexed under their stated mathematical conditions.

Result recovered or investigated	PMM inputs used	Status	Companion reference
R0: Local Lorentz and mass-shell correspondence	Two-vantage reciprocal scalars, r and local light-cone algebra	Algebraic recovery; physical bridge conditional	Recovery R0
R1: Newtonian normalization	Φ-derived source, κ, $\varphi = -\ln \kappa$, Clock Flux, $Q^A = \mathrm{M}$	Newtonian normalization under stated source and coupling identification	Recovery R1
R2: Weak-field temporal GR correspondence	Normalized κ, static lapse, φ and Newtonian potential	Temporal weak-field correspondence; spatial metric bridge separate	Recovery R2
R3: Null and light-sector correspondence	D null propagation, angular progression and reciprocal readouts	Local c and null proper-time correspondence	Recovery R3
R4: Stationary-source accounting	Latest-record source projection and lineage typing	Stationary and Newtonian sum tests under supplied source entries	Recovery R4
R5: Static spherical / Schwarzschild weak-field branch	κ, Clock–Distance Reciprocity, spherical spatial geometry	Spherical recovery branch; general spatial bridge separate	Recovery R5; Core §34.9–34.10
R6: Curved-space null optics	κ, null geometry and spatial metric	Open	Recovery R6
R7: Equivalence-principle correspondence	D clock and relational gravitational response	Open	Recovery R7
R8: Continuum curvature	Discrete δ network and Clock Flux	Open	Recovery R8
R9: Relativistic stress-energy correspondence	A/Φ source, event-to-lineage transfer and D stress	Open	Recovery R9
R10: Lineage interval and relativistic source reconstruction	Angular phase interval, δ embedding, lineage continuity and null-source scale	Conditional / partial source reconstruction	Recovery R10
R11: Causal lineage and source-state accounting	Lineage Lorentz embedding, four-momentum continuity, carried source	Conditional pass; native transfer dynamics separate	Recovery R11
R12: Event-to-lineage transfer	ΔP conservation, elastic contact and null-carrier channels	Partial conservation accounting; dynamical selector open	Recovery R12
R13: Native D interaction / C_2 outcome	D invariants, θ^D, r, a, δ, Φ and admissible channels	Native outcome selection unresolved	Recovery R13
R14: Legacy dynamics compatibility	Angular progress, normalized event rate, lineage momentum ledger	Partial legacy recovery	Recovery R14

Result recovered or investigated	PMM inputs used	Status	Companion reference
R17: Electron energy and angular-action correspondence	Electron rest-energy input, D angular action and length/time scales	Correspondence using supplied electron energy	Recovery R17
Born probability weighting	Reciprocal B-contact, closure-basis invariance and quadratic weighting	Quadratic contact-probability construction under reciprocal symmetry and closure invariance; §5.3 candidate	Core §5.3
A-phase information capacity	A-phase rail area F^{A}_{tu}, $A_{\mathrm{ref}} = 4\ell_p^2$, $q_A = \mathrm{F}^{A}_{tu}/A_{\mathrm{ref}}$	Horizon-normalized information capacity $H_{\max} = q_A/\ln 2$	Core §25.3

Recovery firewall. PMM's first principles define the BDA event architecture. A recovery match establishes compatibility with an established physical result within the stated construction and limit. The recovered theory enters as a comparison standard rather than an adopted PMM first principle. Supporting derivations and tests are cited in the Recovery Companions where available; the Born construction is identified in core §5.3.

Predictions and Falsification Tests

PMM's recovery program establishes compatibility with known results. This section states what PMM asserts that observation could contradict.

Horizon acceleration scale. From the horizon temperature $T_H = \hbar H/(2\pi k_B)$, the A-phase angular rate is $\omega_{A,H} = k_B T_H/\hbar = H/(2\pi)$. The c realization gauge converts that rate to an acceleration:

$$\mathbf{a_0^{PMM}(t)} \;=\; \mathbf{cH(t)/(2\pi)}$$

No fitted coefficient enters the chain. For $H_0 = 67.4$ km/s/Mpc this gives $a_0 \approx 1.04 \times 10^{-10}\ m/s^2$; for $H_0 = 73.0$, $a_0 \approx 1.13 \times 10^{-10}\ m/s^2$. Galaxy-scale fits adopt $a_{0,\mathrm{obs}} \approx 1.16$–$1.20 \times 10^{-10}\ m/s^2$.

Redshift discriminator. The Hubble-horizon branch predicts $a_0^{\mathrm{PMM}}(z)/a_0^{\mathrm{PMM}}(0) = H(z)/H_0$. A constant-$\Lambda$ horizon branch instead gives a ratio of 1 at all z. These two branches are observationally distinguishable and neither admits a free parameter.

Falsification condition. Define the residual diagnostic $C_{a,\mathrm{obs}} = [2\pi a_{0,\mathrm{obs}}/(cH)]^2$. Exact agreement gives $C_{a,\mathrm{obs}} = 1$. A statistically robust departure from 1, or an observed $a_0(z)$ scaling inconsistent with both branches, falsifies the chain rather than being absorbed into it.

Architectural status. The horizon-temperature readout gives $\omega_{A,H} = k_B T_H/\hbar = H/(2\pi)$, while P7 fixes $\omega^A = E^D/\hbar$. Reconciling these rates, or deriving a_0 directly from π-closure, is an open derivation target. The falsification test is stated independently: the measured ratio either tracks H(z) or it does not.

Open derivations

reconcile the horizon-temperature A-phase angular rate $\omega_{A,H} = H/(2\pi)$ with P7 ($\omega^A = \omega^D$), or derive a_0^{PMM} from the π-closure architecture directly.

derive or reject a native metric/measure and measurable $C_1/C_2/C_3$ consequence for the candidate Action Cube F^{s}_{tu} and Q(P), including P = 1 degeneracy, $\mathrm{P} \to 0^+$ domain behavior and the role of Θ_{win}; preserve independent closure counting and the J · s Action Ledger $\mathrm{F}^{\mathrm{led}}_{tu}$

Electron correspondence boundary. Recovery R17 evaluates existing angular-action and scale readouts using the supplied electron rest energy. It does not derive electron mass or identify an observed FTU closure. An independent determination of that energy and an operational event-boundary identification remain separate targets.

native C_3 selection or rejection of $A_{\mathrm{cl}} = A_G$ at the gravitational-crease limiting boundary

native active source-extraction map $\mathcal{S}[\Phi]$ from persistent record history to current carried source s^A, including the event-to-lineage energy-transfer / μ update law

full completed-record gravitational source Q^A and its relation to $\mathcal{S}[\Phi]$

derive or reject the covariant Einstein coupling from endpoint-record-sourced Clock Flux together with the $\kappa \to$ metric-rate bridge, including whether the standard coefficient $8\pi G_N/c^4$ follows without duplicating one shared event's source energy or introducing a c^4 FTU grade; the two endpoint records are not to be counted as two independent source energies

native value of λ_G, if derivable beyond accepting G_N

native D interaction-energy and stress derivation

A-side stress-energy transcription

native derivation or rejection of the event-edge bridge $d\alpha \leftrightarrow \delta_{ij}$, and any deeper $z^D \leftrightarrow \delta$ relation if one exists

native temporal-state-space metric / inner products among T^B, T^D, and T^A; temporal-cube orthogonality if derivable

native derivation or rejection of the spherical Clock–Distance Reciprocity branch and the static clock–curvature recovery candidate

nonlinear transverse/shear dynamics for L, a, and cross-FTU principal orientation; test against the quadratic static-vacuum residual

full SU(2) generator set from PMM-native D geometry, or proof that the planar SO(2)-type half-angle transporter generated by J is the maximal native result

cross-FTU orientation persistence through $C_3/A/\Phi/C_1/C_2$, including whether the minimal A record already carries sufficient information

angular temporal recursion: derive or reject the native $\Phi_{\prec\beta} \to$ B preparation law $\mathcal{G}\theta$, including relative-phase persistence for coherent interference and intensity-only replay for ordinary video

derive or reject retention or reconstruction of original ψ^B, relative phase, or other data beyond R^A, using an explicit C_3/Φ specification; existing scale/shape and phase-span reconstructions do not establish these stronger properties

derive or reject an antisymmetric inter-participant polarity $P_{ij} = -P_{ji}$ distinct from the internal c^1/c^2 imbalance r

physical status and falsification test for candidate 4π relational transport

full nonlinear M_{op} evolution law for pairwise δ-network motion, acceleration/deceleration, gravity coupling, and transport in time

many-FTU coarse-graining / continuum limit from the discrete relational network

covariant continuum limit

independent predictions beyond recovery

The gravity derivation concentrates on the completed-record source, native clock response, stress transcription, and continuum limit.

37. Research Continuation

Research priorities are listed in Open derivations.